Objective Questions

1. The surge impedance of 50 miles long underground cable is 50 ohms. For a 25 miles length it will be:

   (*a*) 25 ohms (*b*) 50 ohms

   (*c*) 100 ohms (*d*) None of the above.

2. For a load flow solution the quantities normally specified at a voltage controlled bus are:

   (*a*) P and Q (*b*) P and $|V|$

   (*c*) Q and $|V|$ (*d*) P and δ.

3. For reducing tower footing resistance it is better to use:

   (*a*) Chemical and ground rods only

   (*b*) Chemical and counterpoise only

   (*c*) Ground rod and counterpoise only

   (*d*) Chemical, ground rods and counterpoise.

4. The positive, negative and zero sequence impedances of a solidly grounded system under steady state condition always follow the relations:

   (*a*) $Z_1 > Z_2 > Z_0$ (*b*) $Z_1 < Z_2 < Z_0$

   (*c*) $Z_0 < Z_1 < Z_2$ (*d*) None of the above.

5. Mho relay is normally used for the protection of:

   (*a*) Long transmission lines (*b*) Medium length lines

   (*c*) Short length lines (*d*) No length criterion.

6. Reactance relay is normally preferred for protection against:

   (*a*) Earth faults only (*b*) Phase faults only

   (*c*) None of the above.

7. If the fault current is 2000 amps, the relay setting 50% and the C.T. ratio is 400/5, then the plug setting multiplier will be:

   (*a*) 25 amps (*b*) 15 amps

   (*c*) 50 amps (*d*) None of the above.

8. Carrier current protection scheme is normally used for:
(*a*) HV transmission lines only
(*b*) HV cables only
(*c*) HV transmission and cables.

9. The ratio of reset to pick up current for an induction cup relay is approx:
(*a*) 0.99 (*b*) 1.01
(*c*) 0.75 (*d*) None of the above.

10. The rotation of disc of an induction disc relay under the poles is:
(*a*) From unshaded pole to shaded pole
(*b*) From shaded pole to unshaded pole
(*c*) It depends upon the magnitude of current
(*d*) It depends upon the C.T. secondary connection.

11. If the time of operation of a relay for unity TMS is 10 secs., the time of operation for 0.5 TMS will be:
(*a*) 20 secs (*b*) 5 secs
(*c*) 10 secs (*d*) None of the above.

12. The shape of the disc of an induction disc relay is:
(*a*) Circular (*b*) Spiral
(*c*) Elliptic.

13. If the phase angle of the voltage coil of a directional relay is 50° the maximum torque angle of the relay is:
(*a*) 130° (*b*) 100°
(*c*) 25° (*d*) None of the above.

14. A reactance relay is:
(*a*) Voltage restrained directional relay
(*b*) Directional restrained over-current relay
(*c*) Voltage restrained over-current relay
(*d*) None of the above.

15. The per cent bias for a generator protection lies between:
(*a*) 5 to 10 (*b*) 10 to 15
(*c*) 15 to 20 (*d*) None of the above.

16. For protection of parallel feeders fed from one end the relays required are:
(*a*) Non-directional relays at the source end and directional relays at the load end.
(*b*) Non-directional relays at both the ends.
(*c*) Directional relays at the source end and non-directional at the load end.
(*d*) Directional relays at both the ends.

17. The number of pilot wires required for protecting 3-phase transmission lines using Translay system of protection is:

(*a*) 6 (*b*) 4

(*c*) 3 (*d*) 2.

18. A shunt fault is characterized by:

(*a*) Increase in current, frequency and p.f.

(*b*) Increase in current reduction in frequency and p.f.

(*c*) Increase in current and frequency but reduction in p.f.

(*d*) None of the above.

19. For measuring positive, negative and zero sequence voltages in a system, the reference is taken as:

(*a*) Neutral of the system only

(*b*) Ground only

(*c*) For zero sequence neutral and for positive and negative the ground

(*d*) None of the above.

20. The most economic load on an overhead line is:

(*a*) Greater than the natural load (*b*) Less than the natural load

(*c*) Equal to the natural load (*d*) None of the above is necessary.

21. The most economic load on an underground cable is:

(*a*) Greater than its surge loading (*b*) Less than the surge loading

(*c*) Equal to the surge loading (*d*) None of the above is necessary.

22. For a long transmission line, for a particular receiving end voltage, when sending end voltage is calculated, it is more than the actual value when calculated by:

(*a*) Load end capacitance method (*b*) Nominal T method

(*c*) Nominal π method (*d*) None of the above methods.

23. The p.u. impedance value of an alternator corresponding to base values 13.2 kV and 30 MVA is 0.2 p.u. The p.u. value for the base values 13.8 kV and 50 MVA will be:

(*a*) 0.306 p.u. (*b*) 0.33 p.u.

(*c*) 0.318 p.u. (*d*) 0.328 p.u.

24. For the same rupturing capacity, the actual current to be interrupted by an HRC fuse is:

(*a*) Much less than any CB (*b*) Much more than any CB

(*c*) Equal to the CB (*d*) None of the above is necessary.

25. The stability of arc in vacuum depends upon:

(*a*) The contact material only

(*b*) The contact material and its vapour pressure

(*c*) The circuit parameters only

(*d*) The combination of (*b*) and (*c*).

26. To limit current chopping in vacuum circuit breakers, the contact material used has:
(*a*) High vapour pressure and low conductivity properties
(*b*) High vapour pressure and high conductivity properties
(*c*) Low vapour pressure and high conductivity properties
(*d*) Low vapour pressure and low conductivity properties.

27. The capacitor switching is easily done with:
(*a*) Air blast circuit breaker (*b*) Oil C.B.
(*c*) Vacuum C.B. (*d*) Any one of the above.

28. Where voltages are high and current to be interrupted is low the breaker preferred is:
(*a*) Air blast C.B. (*b*) Oil C.B.
(*c*) Vacuum C.B. (*d*) Any one of the above.

29. For rural electrification in a country like India with complex network it is preferable to use:
(*a*) Air break C.B. (*b*) Oil C.B.
(*c*) Vacuum C.B. (*d*) M.O. C.B.

30. The current chopping tendency is minimised by using the SF_6 gas at relatively:
(*a*) High pressure and low velocity (*b*) High pressure and high velocity
(*c*) Low pressure and low velocity (*d*) Low pressure and high velocity.

31. SF_6 gas has excellent heat transfer properties because of its:
(*a*) Higher molecular weight (*b*) Low gaseous viscosity
(*c*) Higher dielectric strength (*d*) A combination of (*a*) and (*b*)
(*e*) A combination of (*b*) and (*c*).

32. A fault is more severe from the view point of RRRV if it is a:
(*a*) Short line fault (*b*) Medium length line fault
(*c*) Long line fault (*d*) None of the above.

33. The most suitable C.B. for short line fault without switching resistor is:
(*a*) Air blast C.B. (*b*) M.O.C.B.
(*c*) SF_6 Breaker (*d*) None of the above.

34. The voltages at the two ends of a line are 132 kV and its reactance is 40 ohms. The capacity of the line is:
(*a*) 435.6 MW (*b*) 217.5 MW
(*c*) 251.5 MW (*d*) 500 MW.

35. An overhead line with series compensation is protected using:
(*a*) Impedance relay (*b*) Reactance relay
(*c*) Mho relay (*d*) None of the above.

36. The typical value of SCR for modern alternators is:
(*a*) 1.5 (*b*) 1.2
(*c*) 1.0 (*d*) 0.5.

37. In case of a large size alternator the voltage control becomes serious if:

(*a*) SCR is high (*b*) SCR is low

(*c*) Voltage control is independent of SCR.

38. A large-size alternator is protected against overloads by providing:

(*a*) Overcurrent relays (*b*) Temperature sensitive relays

(*c*) Thermal relays (*d*) A combination of (*a*) and (*b*)

(*e*) A combination of (*b*) and (*c*).

39. The impulse ratio of a rod gap is:

(*a*) Unity (*b*) Between 1.2 and 1.5

(*c*) Between 1.6 and 1.8 (*d*) Between 2 and 2.2.

40. Shunt compensation in an EHV is resorted to:

(*a*) Improve the stability

(*b*) Reduce the fault level

(*c*) Improve the voltage profile

(*d*) As a substitute for synchronous phase modifier.

41. In order to have lower cost of electrical energy generation:

(*a*) The load factor and diversity factor should be low

(*b*) The load factor should be low but diversity factor should be high

(*c*) The load factor should be high but diversity factor low

(*d*) The load factor and diversity factors should be high.

42. The insulation of the modern EHV lines is designed based on:

(*a*) The lighting voltage (*b*) The switching voltage

(*c*) Corona (*d*) RI.

43. The size of conductor on modern EHV lines is obtained based on:

(*a*) Voltage drop (*b*) Current density

(*c*) Corona (*d*) (*a*) and (*b*).

44. A Mho relay is a:

(*a*) Voltage restrained directional relay

(*b*) Voltage controlled over current relay

(*c*) Directional restrained over current relay

(*d*) Directional restrained over voltage relay.

45. For stability and economic reasons we operate the transmission line with power angle in the range:

(*a*) 10° to 25° (*b*) 30° to 45°

(*c*) 60° to 75° (*d*) 65° to 80°.

46. High voltage d.c. testing for HV machines is resorted because:

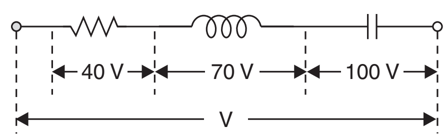
(*a*) Certain conclusions regarding the continuous ageing of an insulation can be drawn.

(*b*) The stress distribution is a representation of the service condition
(*c*) Standardization on the magnitude of voltage to be applied is available
(*d*) The stresses do not damage the coil end insulation.

47. For a lumped inductive load, with increase in supply ferquency:
(*a*) P and Q increase
(*b*) P increases, Q decreases
(*c*) P decreases, Q increases
(*d*) P and Q decrease.

48. Standard impulse testing of a power transformer requires:
(*a*) Two applications of chopped wave followed by one application of a full wave
(*b*) One application of chopped wave followed by one application of a full wave
(*c*) One application of chopped wave followed by two applications of a full wave
(*d*) None of the above.

49. The supply voltage $|V|$ in diagram below is:



(*a*) 210 V
(*b*) 70 V
(*c*) 50 V
(*d*) 230 V.

50. The chances of arc interruption in subsequent current zero:
(*a*) Increases in case of *OCB*
(*b*) Decreases in case of *OCB*
(*c*) Interruption is always at first current zero (in *OCB*)
(*d*) None of the above.

51. The chances of arc interruption in subsequent current zeros:
(*a*) Increases in case of *ABCB* but decreases in *OCB*
(*b*) Decreases in case of *ABCB* but increases in *OCB*
(*c*) Decreases in both the cases
(*d*) Increases in both the cases.

52. The velocity of travelling wave through a cable of relative permittivity 9 is:
(*a*) 9×10^8 metres/sec
(*b*) 3×10^8 metres/sec
(*c*) 10^8 metres/sec
(*d*) 2×10^8 metres/sec.

53. An overhead line with surge impedance 400 ohms is terminated through a resistance R. A surge travelling over the line does not suffer any reflection at the junction if the value of R is:
(*a*) 20 ohms
(*b*) 200 ohms
(*c*) 800 ohms
(*d*) None of the above.

54. The rate of rise of restriking voltage depends upon:
(*a*) The type of circuit breaker
(*b*) The inductance of the system only
(*c*) The capacitance of the system only
(*d*) The inductance and capacitance of the system.

55. Gas turbines can be brought to the bus bar from cold in about:
(*a*) 2 minutes (*b*) 30 minutes
(*c*) 1 Hour (*d*) 2 Hours.

56. A 3-phase breaker is rated at 2000 MVA, 33 kV, its making current will be:
(*a*) 35 kA (*b*) 49 kA
(*c*) 70 kA (*d*) 89 kA.

57. If a combination of HRC fuse and circuit breaker is used, the C.B. operates for:
(*a*) Low overload currents (*b*) Short circuit current
(*c*) Under all abnormal current (*d*) The combination is never used in practice.

58. The impulse ratio of a gap of given geometry and dimension is:
(*a*) Greater with solid than with air dielectric
(*b*) Greater with air than with solid dielectric
(*c*) Same for both solid and air dielectric.

59. Tick out the correct one:
(*a*) The insulators and lightning arresters should have high impulse ratio
(*b*) The insulators and lightning arresters should have low impulse ratio
(*c*) The insulator should have high impulse ratio and lightning arrester low
(*d*) The lightning arrester should have high impulse ratio but insulator low.

60. Phase modifier is normally installed in the case of:
(*a*) Short transmission lines (*b*) Medium length lines
(*c*) Long length lines (*d*) For all length lines.

61. Series compensation on EHV lines is resorted to:
(*a*) Improve the stability
(*b*) Reduce the fault level
(*c*) Improve the voltage profile
(*d*) As a substitute for synchronous phase modifier.

62. Ferranti effect on long overhead lines is experienced when it is:
(*a*) Lightly loaded (*b*) On full load at unity p.f.
(*c*) On full load at 0.8 p.f. lag (*d*) In all these cases.

63. Stringing chart is useful for:
(*a*) Finding the sag in the conductor (*b*) In the design of tower
(*c*) In the design of insulator string (*d*) Finding the distance between the tower.

64. The presence of earth in case of overhead lines:

(*a*) Increases the capacitance (*b*) Increases the inductance

(*c*) Decreases the capacitance (*d*) Decreases the inductance.

65. For an existing a.c. transmission line the string efficiency is 80%. Now if d.c. voltage is supplied for the same set up, the string efficiency will be:

(*a*) 80% (*b*) More than 80%

(*c*) Less than 80% (*d*) 100%.

66. Effect of increase in temperature in overhead transmission lines is to:

(*a*) Increase the stress and length

(*b*) Decrease the stress and length

(*c*) Decrease the stress but increase the length

(*d*) None of the above.

67. The Buchholz relay protects a transformer from:

(*a*) All types of internal fault (*b*) A turn to turn fault

(*c*) Winding to winding fault (*d*) None of them.

68. The Self GMD method is used to evaluate:

(*a*) Inductance (*b*) Capacitance

(*c*) Inductance and capacitance both (*d*) None of the above

of the overhead transmission lines.

69. A voltage-controlled bus is treated as a load bus in subsequent iteration when its:

(*a*) Voltage limit is violated (*b*) Active power limit is violated

(*c*) Reactive power limit is violated (*d*) Phase angle limit is violated.

70. The voltage of a particular bus can be controlled by controlling the:

(*a*) Phase angle (*b*) Reactive power of the bus

(*c*) Active power of the bus (*d*) Phase angle and reactive power.

71. In case of a 3-phase short circuit in a system, the power fed into the system is:

(*a*) Mostly reactive (*b*) Mostly active

(*c*) Active and reactive both equal (*d*) Reactive only.

72. The load flow solution is always assured in case of:

(*a*) Newton-Raphson method (*b*) Gauss method

(*c*) Gauss-Seidel method (*d*) None of these methods guarantees.

73. If α is the angle of voltage wave at which an R-L circuit is switched in and θ is the impedance angle of the R-L circuit, there will be no transient when the circuit is switched in, if:

(*a*) $\alpha = \theta$ (*b*) $\alpha = 90 - \theta$

(*c*) $\alpha = 90 + \theta$ (*d*) None of the above.

74. At a particular unbalanced node, the real powers specified are:

Leaving the node 20 MW, 25 MW

Entering the node 60 MW, 30 MW

The balancing power will be:

(*a*) 30 MW leaving the node

(*b*) 45 MW leaving the node

(*c*) 45 MW entering the node

(*d*) 22.5 MW entering the node and 22.5 MW leaving the node.

75. The loss coefficients for a two-bus system are given as follows:

(*a*) $B_{11} = 0.02$, $B_{22} = 0.05$, $B_{12} = 0.01$, $B_{21} = 0.015$.

(*b*) $B_{11} = 0.02$, $B_{22} = 0.04$, $B_{12} = -0.01$, $B_{21} = 0.001$.

(*c*) $B_{11} = 0.03$, $B_{22} = 0.005$, $B_{12} = -0.001$, $B_{21} = -0.001$.

(*d*) $B_{11} = 0.03$, $B_{22} = 0.05$, $B_{12} = 0.001$, $B_{21} = -0.001$.

76. For economic operation, the generator with highest positive incremental transmission loss will operate at:

(*a*) The lowest positive incremental cost of production

(*b*) The lowest negative incremental cost of production

(*c*) The highest positive incremental cost of production

(*d*) None of the above.

77. The inertia constant H of a machine of 200 MVA is 2 p.u. its value corresponding to 400 MVA will be:

(*a*) 4.0 (*b*) 2.0

(*c*) 1.0 (*d*) 0.5.

78. The inertia constant of two groups of machines, which swing together are M_1 and M_2. The inertia constant of the system is:

(*a*) $\dfrac{M_1M_2}{M_1 + M_2}$ (*b*) $M_1 - M_2$, $M_1 > M_2$

(*c*) $M_1 + M_2$ (*d*) $\dfrac{M_1 + M_2}{M_1M_2}$.

79. The leakage resistance of a 50 km long cable is 1 MΩ. For a 100 km long cable it will be:

(*a*) 1 MΩ (*b*) 2 MΩ

(*c*) 0.66 MΩ (*d*) None of the above.

80. The coefficient of reflection for current for an open ended line is:

(*a*) 1.0 (*b*) 0.5

(*c*) – 1.0 (*d*) Zero.

81. The connected load of a consumer is 2 kW and his maximum demand is 1.5 kW. The load factor of the consumer is:

(*a*) 0.75 (*b*) 0.375

(*c*) 1.33 (*d*) None of the above.

82. The maximum demand of a consumer is 2 kW and his daily energy consumption is 20 units. His load factor is:

(*a*) 10% (*b*) 41.6%

(*c*) 50% (*d*) None of the above.

83. Resistance switching is normally resorted in case of:

(*a*) Bulk oil circuit breakers (*b*) Minimum oil circuit breakers

(*c*) Air blast circuit breakers (*d*) All types of breakers.

84. If the inductance and capacitance of a system are 1.0 H and 0.01 μF respectively and the instantaneous value of current interrupted is 10 amp, the voltage across the breaker contacts will be:

(*a*) 50 kV (*b*) 100 kV

(*c*) 60 kV (*d*) 57 kV.

85. The p.u. synchronous impedance of a machine is 2.0 p.u., its SCR is:

(*a*) 2 (*b*) 1.414

(*c*) 0.5 (*d*) 0.707.

86. Sheaths are used in cables to:

(*a*) Provide proper insulation (*b*) Provide mechanical strength

(*c*) Prevent ingress of moisture (*d*) None of the above.

87. Tick out the correct statement:

(*a*) The higher the initial load, the larger the critical clearing angle

(*b*) The higher the initial load, the lower the critical clearing angle

(*c*) The initial load has nothing to do with the critical clearing angle

(*d*) The higher the operating time of the circuit breaker, the larger will be the critical clearing angle.

88. The capacitance and inductance per unit length of a line operating at 110 kV are 0.01 μF and 2 mH. The surge impedance loading of the line is:

(*a*) 40 MVA (*b*) 30 MVA

(*c*) 27 MVA (*d*) None of the above.

89. An RLC series circuit remains predominantly inductive:

(*a*) At resonance frequency (*b*) Below resonance frequency

(*c*) Above resonance frequency (*d*) At the lower half power frequency.

90. An alternator is delivering a load current; its per cent regulation is found to be zero. The type of load it delivers is:

(*a*) Capacitive (*b*) Inductive

(*c*) Resistive.

91. A short length of cable between the dead end tower and the power transformer :

(*a*) Always reduces the steepness of the incident wave (*b*) Reduces the steepness of the wave under certain conditions only

(*c*) It does not change the steepness.

92. For protection of rotating machines against lightning surges a combination of:

(*a*) Lightning conductor and capacitor

(*b*) Lightning conductor and lightning arrester

(*c*) Lightning arrester and capacitor

(*d*) Lightning arrester alone is used.

93. An alternator is delivering a balanced load at unity p.f. the phase angle between line voltage and line current is:

(*a*) 90° (*b*) 60°

(*c*) 30° (*d*) 0°.

94. For the stable operation of interconnected system, the passive element that can be used a interconnecting element is:

(*a*) Reactor (*b*) Resistor

(*c*) Capacitor (*d*) Resistor and capacitor.

95. If two synchronous generators are connected, loss of synchronism will result in:

(*a*) Stalling of generators

(*b*) Wild fluctuations in current

(*c*) Wild fluctuations in current and voltage

(*d*) None of the above.

96. A generator is connected to a synchronous motor. From stability point of view it is preferable to have:

(*a*) Generator neutral reactance grounded and motor neutral resistance grounded

(*b*) Generator and motor neutrals resistance grounded

(*c*) Generator and motor neutrals reactance grounded

(*d*) Generator neutral resistance and motor neutral reactance grounded.

97. A 50 Hz, four-pole turboalternator rated at 20 MVA, 13.2 kV has an inertia constant $H = 4$ kW sec/kVA. The K.E. stored in the rotor at synchronous speed is:

(*a*) 80 kilojoules

(*b*) 80 megajoules

(*c*) 40 megajoules

(*d*) 20 megajoules.

98. The breakdown strength of air at STP is 21 kV/cm. Its breakdown strength at 30°C and 72 cm of Hg will be:

(*a*) 21.25 kV/cm (*b*) 20.2 kV/cm

(*c*) 23 kV (*d*) 19.5 kV/cm.

99. Corona loss is less when the shape of the conductor is:

(*a*) Circular
(*b*) Flat
(*c*) Oval
(*d*) Independent of shape.

100. Corona loss increases with:

(*a*) Increase in supply frequency and conductor size
(*b*) Increase in supply frequency but reduction in conductor size
(*c*) Decrease in supply frequency and conductor size
(*d*) Decrease in supply frequency but increase in conductor size.

101. The corona loss on a particular system at 50 Hz is 1 kW/phase per km. The corona loss on the same system with supply frequency 25 Hz will be:

(*a*) 1 kW/phase/km
(*b*) 0.5 kW/phase/km
(*c*) 0.667 kw/phase/km
(*d*) None of the above.

102. A system is said to be effectively grounded if its:

(*a*) Neutral is grounded directly
(*b*) Ratio of $\frac{X_0}{X_1} > 3.0$
(*c*) Ratio of $\frac{R_0}{X_1} > 2.0$
(*d*) Ratio of $\frac{X_0}{X_1} < 3.0$.

103. For effective application of counterpoise is should be buried into the ground to a depth of:

(*a*) 1 metre
(*b*) 2 metres
(*c*) Just enough to avoid theft
(*d*) None of the above.

104. If r is the radius of the conductor and R the radius of the sheath of the cable, the cable operates stably from the view point of dielectric strength if:

(*a*) $\frac{r}{R} > 1.0$
(*b*) $\frac{r}{R} < 1.0$
(*c*) $\frac{r}{R} < 0.632$
(*d*) $\frac{r}{R} < 0.368$.

105. Three insulating materials with same maximum working stress and permittivities 2.5, 3.0, 4.0 are used in a single core cable. The location of the materials with respect to the core of the cable will be:

(*a*) 2.5, 3.0, 4.0
(*b*) 3.0, 2.5, 4.0
(*c*) 4.0, 3.0, 2.5
(*d*) 4.0, 2.5, 3.0.

106. Three insulating materials with breakdown strengths of 250 kV/cm, 200 kV/cm, 150 kV/cm and permittivities of 2.5, 3.0 and 3.5 are used in a single core cable. If the factor of safety for the materials is 5, the location of the materials with respect to the core of the cable will be:

(*a*) 2.5, 3.0, 3.5
(*b*) 3.0, 2.5, 3.5
(*c*) 3.5, 3.0, 2.5
(*d*) 3.5, 2.5, 3.0.

107. The coefficient of reflection of voltage for a short circuited line is:

(*a*) 1.0 (*b*) – 1.0

(*c*) 0 (*d*) 2.0.

108. The insulation resistance of a cable of length 10 km is 1 MΩ, its resistance for 50 km length will be:

(*a*) 1 MΩ (*b*) 5MΩ

(*c*) 0.2 MΩ (*d*) None of the above.

109. The capacitance of a 3-core cable between any two conductors with sheath earthed is 2 μF. The capacitance per phase will be:

(*a*) 1 μF (*b*) 4 μF

(*c*) 0.667 μF (*d*) 1.414 μF.

110. If δ is the loss angle of the cable, its power factor is:

(*a*) sin δ

(*b*) cos δ

(*c*) Power factor is independent of δ

(*d*) Power factor depends upon δ but it is not as per (*a*) and (*b*).

111. The effect of bonding the cable is:

(*a*) To increase the effective resistance and inductance

(*b*) To increase the effective resistance but reduce inductance

(*c*) To decrease the effective resistance and inductance

(*d*) To decrease the effective resistance but increase the inductance.

112. In order to eliminate sheath losses, a successful method is:

(*a*) To transpose the cable along with cross bonding

(*b*) Transpose the cables only

(*c*) Cross bonding the cables is enough

(*d*) None of the above is effective.

113. The sending end voltage of a feeder with reactance 0.2 p.u. is 1.2 p.u. If the reactive power supplied at the receiving end of the feeder is 0.3 p.u., the approximate drop of volts in the feeder is:

(*a*) 0.2 p.u. (*b*) 0.06 p.u.

(*c*) 0.05 p.u. (*d*) 0.072 p.u.

114. If a synchronous machine is overexcited, it takes lagging vars from the system when it is operated as a:

(*a*) Synchronous motor

(*b*) Synchronous generator

(*c*) Synchronous motor as well as generator

(*d*) None of the above.

115. If a synchronous machine is underexcited it takes lagging vars from the system when it is operated as a:

(*a*) Synchronous motor

(*b*) Synchronous generator

(*c*) Synchronous motor as well as generator

(*d*) None of the above.

116. A synchronous machine has higher capacity for:

(*a*) Leading p.f.

(*b*) Lagging p.f.

(*c*) It does not depend upon the p.f. of the machine

(*d*) It depends upon the p.f. of the load.

117. A machine designed to operate at full load is physically heavier and is costlier if the operating p.f. is:

(*a*) Lagging (*b*) Leading

(*c*) The size and cost do not depend on p.f.

118. For the same voltage boost, the reactive power capacity is more for a:

(*a*) Shunt capacitor (*b*) Series capacitor

(*c*) It is same for both series and shunt.

119. A synchronous phase modifier as compared to a synchronous motor used for mechanical loads has:

(*a*) Larger shaft and higher speed (*b*) Smaller shaft and higher speed

(*c*) Larger shaft and smaller speed (*d*) Smaller shaft and smaller speed.

120. If I_{a_1} is the positive sequence current of an alternator and Z_1, Z_2 and Z_0 are the sequence impedances of the alternator. The drop produced by the current I_{a_1} will be:

(*a*) $I_{a_1} Z_1$ (*b*) $I_{a_1}(Z_1 + Z_2)$

(*c*) $I_{a_1}(Z_1 + Z_2 + Z_0)$ (*d*) $I_{a_1}(Z_2 + Z_0)$.

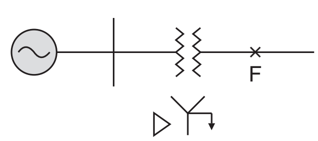
121. For the system shown in diagram below, a line-to-ground fault on the line side of the transformer is equivalent to:

(*a*) A line-to-ground fault on the generator side of the transformer

(*b*) A line-to-line fault on the generator side of the transformer

(*c*) A double line-to-ground on the generator side of the transformer

(*d*) A 3-phase fault on the generator side of the transformer.



122. The positive sequence component of voltage at the point of fault is zero when it is a:

(*a*) 3-phase fault (*b*) *L-L* fault

(*c*) *L-L-G* fault (*d*) *L-G* fault.

123. Tick out the correct statement:

(*a*) The negative and zero sequence voltages are maximum at the fault point and decrease towards the neutral

(*b*) The negative and zero sequence voltages are minimum at the fault point and increase towards the neutral

(*c*) The negative sequence is maximum and zero sequence minimum at the fault point and decrease and increase respectively towards the neutral.

(*d*) None of the above.

124. For complete protection of a 3-phase line:

(*a*) Three-phase and three-earth fault relays are required

(*b*) Three-phase and two-earth fault relays are required

(*c*) Two-phase and two-earth fault relays are required

(*d*) Two-phase and one-earth fault relays are required.

125. The phase comparators in case of static relays and electro-mechanical relays normally are:

(*a*) Sine and cosine comparators respectively

(*b*) Cosine and sine comparators respectively

(*c*) Both are cosine comparators

(*d*) Both are sine comparators.

126. The order of the lightning discharge current is:

(*a*) 10,000 amp (*b*) 100 amp

(*c*) 1 amp (*d*) 1 microampere.

127. The magnetising-inrush-current in a transformer is rich in:

(*a*) 3rd harmonics (*b*) 5th harmonics

(*c*) 7th harmonics (*d*) 2nd harmonics.

128. For effective use of a counterpoise wire:

(*a*) Its leakage resistance should be greater than the surge impedance

(*b*) Its leakage resistance should be less than the surge impedance

(*c*) Its leakage resistance should be equal to the surge impedance

(*d*) The two resistances may have any relation.

129. The inertia constants of two groups of machines which do not swing together are M_1 and M_2. The equivalent inertia constant of the system is:

(*a*) $M_1 + M_2$ (*b*) $M_1 - M_2$ if $M_1 > M_2$

(*c*) $\dfrac{M_1 M_2}{M_1 + M_2}$ (*d*) $\sqrt{M_1 M_2}$.

130. If the inductance and capacitance of a system are 1 H and 0.01 μF respectively and the instantaneous value of current interrupted is 10 amps, the value of shunt resistance across the breaker for critical damping is:

(*a*) 100 kΩ (*b*) 10 kΩ
(*c*) 5 kΩ (*d*) 1 kΩ.

131. For load flow solution the quantities specified at a load bus are:

(*a*) P and Q (*b*) P and $|V|$
(*c*) Q and $|V|$ (*d*) P and δ.

132. The solution of coordination equations takes into account:

(*a*) All the system constraints
(*b*) All the operational constraints
(*c*) All the system and operation constraints
(*d*) None of the above.

133. For a two-bus system if the change in load at bus 2 is 5 MW and the corresponding change in generation at bus 1 is 8 MW, the penalty factor of bus 1 is:

(*a*) 0.6 (*b*) 1.67
(*c*) 0.625 (*d*) None of the above.

134. If the penalty factor for bus 1 in a two-bus system is 1.25 and if the incremental cost of production at bus 1 is Rs. 200 per MWhr, the cost of received power at bus 2 is:

(*a*) Rs. 250/MWhr (*b*) Rs. 62.5/MWhr
(*c*) Rs. 160/MWhr (*d*) None of the above.

135. The cost of generation is theoretically minimum if:

(*a*) The system constraints are considered
(*b*) The operational constraints are considered
(*c*) (*a*) and (*b*)
(*d*) The constraints are not considered.

136. The incremental transmission loss of a plant is:

(*a*) Positive always (*b*) Negative always
(*c*) Can be positive or negative.

137. If P_m is the maximum power transferred, the loss on the system is:

(*a*) $\dfrac{P_m}{4}$ (*b*) $\dfrac{P_m}{2}$
(*c*) $\dfrac{3P_m}{4}$ (*d*) None of the above.

138. If X is the system reactance and R its resistance, the power transferred is maximum when:

(*a*) $X = R$ (*b*) $X = \sqrt{2R}$
(*c*) $X = \sqrt{3R}$ (*d*) $X = 2R$.

139. If the inertia constant H of a machine of 200 MVA is 2 p.u. its value corresponding to 400 MVA will be:

(*a*) 4 p.u (*b*) 2 p.u.

(*c*) 1.0 p.u. (*d*) 0.5 p.u.

140. The inertia constant of two groups of machines which do not swing together are M_1 and M_2 such that $M_1 > M_2$. It is proposed to add some inertia to one of the two groups of machines for improving the transient stability of the system. It should be added to:

(*a*) M_1 (*b*) M_2

(*c*) It does not matter whether to add to M_1 or M_2.

141. Tick out the correct one:

(*a*) Higher the SCR of a machine the heavier is its rotor

(*b*) Higher the SCR of a machine the lighter is its rotor

(*c*) The SCR and size of the rotor are not at all related.

142. A 100 V/10 V, 50 VA transformer is converted to 100 V/110 V auto transformer, the rating of the auto transformer is:

(*a*) 550 VA (*b*) 500 VA

(*c*) 110 VA (*d*) 100 VA.

143. A 100 KVA transformer has maximum efficiency of 98% when operating at half full load. Its full load losses are:

(*a*) Cu = 2.04 kW and Iron loss = 0.51 kW

(*b*) Cu = 4.08 kW and Iron loss = 1.02 kW

(*c*) Cu = 3.0 kW and Iron loss = 0.75 kW

(*d*) None of the above.

144. A voltmeter gives 120 oscillations per minute when connected to the rotor. The stator frequency is 50 Hz. The slip of the motor is:

(*a*) 2% (*b*) 4%

(*c*) 5% (*d*) 2.5%.

145. If the excitation of the synchronous generator fails, it acts as a:

(*a*) Synchronous motor (*b*) Synchronous generator

(*c*) Induction motor (*d*) Induction generator.

146. I and T are the line current and the torque respectively when DOL starter is used, these quantities when Y/Δ starter is used, are:

(*a*) $\frac{I}{3}, \frac{T}{\sqrt{3}}$ (*b*) $\frac{I}{\sqrt{3}}, \frac{T}{3}$

(*c*) $\frac{I}{\sqrt{2}}, \frac{T}{\sqrt{2}}$ (*d*) $\frac{I}{2}, \frac{T}{4}$.

147. A 100 KVA transformer has 4% impedance and 50 KVA transformer has 3% impedance. When they are operated in parallel, which transformer will reach full load first.

(*a*) 3% (*b*) 4%

(*c*) The data is insufficient to judge.

148. In EHV transmission lines, efficiency of transmission can be increased by decreasing the corona loss. This is achieved by

(*a*) Increasing the distance between the line conductors

(*b*) Using bundled conductors

(*c*) Using thick conductors

(*d*) Using thin conductors.

149. An alternator having induced emf. of 1.6 p.u. is connected to an infinite bus of 1.0 p.u. If the busbar has reactance of 0.6 p.u. and alternator has reactance of 0.2 p.u., the maximum power that can be transferred is given by

(*a*) 8 p.u. (*b*) 2 p.u.

(*c*) 2.67 p.u. (*d*) 5.0 p.u.

150. If one phase of supply goes off in the case of 3-phase induction motor, the motor

(*a*) Comes to a stop

(*b*) Draws double the initial current and continues to run

(*c*) Draws $1/\sqrt{3}$ times the initial current and continues to run

(*d*) None of the above.

151. Induction generator works between the slip

(*a*) $1 < s < 2$ (*b*) $0.1 < s < 1.0$

(*c*) $s < 0.0$ (*d*) None of the above.

152. The motor which can be used on both a.c. and d.c. is

(*a*) Reluctance motor (*b*) Induction motor

(*c*) d.c. series motor (*d*) None of the above.

153. If transformer frequency is changed from 50 Hz to 60 Hz, the ratio of eddy current loss 50 Hz to 60 Hz at constant voltage is

(*a*) 5/6 (*b*) 25/36

(*c*) 6/5 (*d*) 1.0.

154. An alternator of 300 kW is driven by a prime mover of speed regulation 4% and another alternator of 200 kW driven by a prime mover of speed regulations 3%, the total load they can take is

(*a*) 500 kW (*b*) 567 kW

(*c*) 425 kW (*d*) 257 kW.

155. The minimum oil circuit breaker has less volume of oil because:

(*a*) There is insulation between contacts

(*b*) The oil between the breaker contacts has greater strength

(*c*) Solid insulation is provided for insulating the contacts from earth

(*d*) None of the above is true.

156. Load flow study is carried out for

(*a*) Fault calculations (*b*) Stability studies

(*c*) System planning (*d*) Load frequency control.

157. Arcing on transmission lines is prevented by connecting a suitable

(*a*) Circuit breaker (*b*) Protective relay

(*c*) Inductor in the neutral (*d*) Capacitor in the neutral.

158. The in-rush current of a transformer at no load is maximum if the supply voltage is switched on

(*a*) At zero voltage value (*b*) At peak voltage value

(*c*) At $V/2$ value (*d*) At $\sqrt{3}/2$ V value.

159. If the loading of the line corresponds to the surge impedance loading the voltage at the receiving end is

(*a*) Greater than sending end (*b*) Less than sending end

(*c*) Equal to the sending end (*d*) None of the above is necessary.

160. The voltages of a generator and an infinite bus are given as $0.92\angle 10°$ and $1.0\angle 0°$ respectively. The generator acts as a

(*a*) Shunt coil (*b*) Shunt capacitor

(*c*) The data is insufficient to judge.

161. For a synchronous phase modifier the load angle is

(*a*) 0° (*b*) 25°

(*c*) 30° (*d*) None of the above.

162. The voltages of a generator and an infinite bus are given as $0.92\angle 10°$ and $1.0\angle 0°$ respectively. The active power will flow from

(*a*) Generator to infinite bus (*b*) Infinite bus to generator

(*c*) The data is insufficient to judge.

163. If the penalty factor of a plant is unity, its incremental transmission loss is

(*a*) 1.0 (*b*) – 1.0

(*c*) Zero (*d*) None of the above.

164. In a two plant system, the load is connected at plant no. 2. The loss coefficients

(*a*) B_{11}, B_{12}, B_{22} are nonzero

(*b*) B_{11} and B_{22} are nonzero but B_{12} is zero

(*c*) B_{11} and B_{12} are nonzero but B_{22} is zero

(*d*) B_{11} is nonzero but B_{12} and B_{22} are zero.

165. The diagonal elements of a nodal admittance matrix are strengthened by adding

(*a*) Shunt inductances (*b*) Shunt capacitors

(*c*) Loads (*d*) Generators.

166. The charging reactance of 50 km length of line is 1500 Ω. The charging reactance for 100 km length of line will be

(*a*) 1500 Ω (*b*) 3000 Ω
(*c*) 750 Ω (*d*) 600 Ω.

167. The eddy current loss in the core of a transformer is

(*a*) Inversely proportional to resistivity of the core material
(*b*) Directly proportional to resistivity of the core material
(*c*) Directly proportional to square of resistivity of the core material
(*d*) None of the above is true.

168. The normal practice to specify the making current of a circuit breaker is in terms of

(*a*) r.m.s value (*b*) Peak value
(*c*) Average value (*d*) Both r.m.s. and peak value.

169. A transmission line of 80 km length is operating at 400 Hz, it can be classified as

(*a*) Short length line (*b*) Medium length line
(*c*) Long length line (*d*) all of the above.

170. A transmission line of 210 km length has certain values of parameters A, B, C, D. If the length is made 100 km thus the parameter

(*a*) A increase B decreases (*b*) A decreases B also decreases
(*c*) A and B both increase (*d*) A decreases B increases

171. In the above problem, the parameters

(*a*) C increases D decreases (*b*) C and D both decrease
(*c*) C and D both increase (*d*) C decreases D increases

172. When two transformers of different KVA ratings are connected in parallel they share the load in proportion to their respective KVA rating only when their

(*a*) KVA ratings are identical (*b*) Efficiencies are equal
(*c*) pu impedances are equal (*d*) equivalent impedances are equal.

173. A salient pole synchronous motor is running at no load, its field current is switched off. The motor will

(*a*) come to a stop
(*b*) continue to run at synchronous speed
(*c*) continue to run at a speed slightly less than synchronous speed
(*d*) none of these.

174. In a network the sum of currents entering a node is $5\angle 60°$. The sum of the currents leaving the node is

(*a*) $5 \angle 60°$ (*b*) $5 \angle -60°$
(*c*) $5 \angle 240°$ (*d*) 15 A.

175. In a synchronous machine, in case the axis of field flux is in line with the armature flux, the machine is working

(*a*) as synchronous motor (*b*) as synchronous generator

(*c*) as floating machine (*d*) the m/c will not work.

176. An ideal voltage source is connected across a variable resistance. The variation of current as a function of resistance is given by

(*a*) A straight line passing through the origin

(*b*) a rectangular hyperbola

(*c*) a parabola

(*d*) it could be anyone of the above.

177. Pure inductive circuit takes power (reactive) from the a.c. line when

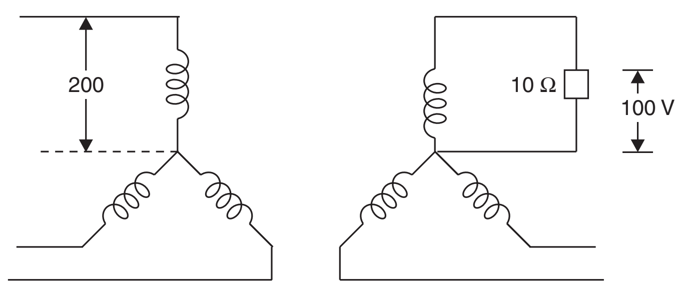
(*a*) both applied voltage and current rise

(*b*) both applied voltage and current decrease

(*c*) applied voltage decreases but current increases

(*d*) (*a*) and (*b*).

178. The current in the primary of the given transformer is



(*a*) 10 *A* (*b*) 5 *A*

(*c*) $10/\sqrt{3}\,A$ (*d*) none of the above.

179. A synchronous machine has its field winding on the stator and armature winding on the rotor. Under steady running condition, the air gap field

(*a*) rotates at synchronous speed w.r.t. stator

(*b*) rotates at synchronous speed in the direction of rotation of the rotor

(*c*) remains stationary w.r.t. stator

(*d*) remains stationary w.r.t. rotor.

180. For a given applied voltage to a transformer, if its magnetic core is replaced by a non-magnetic core

The magnitude of magnetic flux produced by the primary winding

(*a*) remains same (*b*) decreases

(*c*) increases (*d*) all the above alternatives are possible.

181. If a transformer is switched on to a voltage more than the rated voltage, for a certain loading condition (lag p.f.)

(*a*) its p.f. will improve (*b*) its p.f. will deteriorate

(*c*) no efect on p.f. (*d*) it may or may not improve.

182. Three phase transformers which can not be connected in parallel are

(*a*) YY with $\Delta\Delta$ (*b*) $Y\Delta$ with ΔY

(*c*) YY with $Y\Delta$ (*d*) (*b*) and (*c*).

183. For certain geometry and operating voltage of the uncompensated transmission line the ratio of power transfer capability to the surge impedance loading with increase in length

(*a*) increases (*b*) remains unchanged

(*c*) decreases (*d*) no fixed criterion.

184. If the normal system frequency is 50 Hz and if it is operating at 53 Hz, the equipment on the system most adversely affected is

(*a*) Power transformer (*b*) Alternator

(*c*) Turbine (*d*) all the above are equally affected.

185. The operation of the relay which is most affected due to arc resistance is

(*a*) Mho relay (*b*) reactance relay

(*c*) Impedance relay (*d*) all are equally affected.

186. The relay which is most sensitive to power swings (maloperation) is

(*a*) Mho relay (*b*) reactance relay

(*c*) Impedance relay (*d*) all are equally affected.

187. A 100/5 A bar primary current transformer supplies an O/C relay set at 25% pick up and it has a burden of 5 VA. The secondary voltage is

(*a*) 1 volt (*b*) 1.25 volt

(*c*) 2.5 volt (*d*) 4 volt.

188. A 100 : 5 current transformer has secondary winding and lead resistance of 0.2Ω and the secondary burden is 5 VA. The VA output is

(*a*) 5 VA (*b*) 10 VA

(*c*) 7.5 VA (*d*) 2.5 VA.

189. The phase angle error of a current transformer can be reduced by connecting an element across its secondary which is predominantly

(*a*) resistive (*b*) inductive

(*c*) capacitive (*d*) independent of the nature of element.

190. Auto reclosing is used in case of

(*a*) Lightning arrester (*b*) Air blast C.B.

(*c*) Bulk oil C.B. (*d*) Minimum oil C.B.

191. If the normal frequency is 50 Hz, the setting of the under-frequency relay for the most important load could be

(*a*) 49 Hz
(*b*) 48.5 Hz
(*c*) 47.5 Hz
(*d*) there is no grading for under frequency relays.

192. A distance relay is said to be inherently directional if its characteristic on R-X diagram
(*a*) is a straight line off-set from the origin
(*b*) is a circle that passes through the origin
(*c*) is a circle that encloses the origin
(*d*) always a separate directional relay is required.

193. The standard current ratings of an electromagnetic relay are
(*a*) 5 A and 15 A
(*b*) 15 A and 20 A
(*c*) 1 A and 5 A
(*d*) any one of the above.

194. Standard voltage ratings of the electromagnetic relay are
(*a*) 110 V, 63.5 (L-N)
(*b*) 220 V 110 V
(*c*) 230 V 154 V
(*d*) 400 V 231 V.

195. In order to quench the arc quickly and also optimise the dimensions generally
(*a*) oil C.B. is preferred
(*b*) air blast C.B. is preferred
(*c*) SF_6 C.B. is preferred
(*d*) minimum oil C.B. is preferred.

196. The frequency of the carrier in the case of carrier current-pilot scheme is in the range of
(*a*) 1 KHz to 10 KHz
(*b*) 15 KHz to 25 KHz
(*c*) 25 KHz to 50 KHz
(*d*) 50 KHz to 500 KHz.

197. The unit protection scheme provides
(*a*) primary protection
(*b*) Back up protection
(*c*) simultaneous protection
(*d*) remote protection.

198. The $X : R$ ratio for 220 KV line as compared to 400 KV line is
(*a*) greater
(*b*) smaller
(*c*) equal
(*d*) it could be anything.

199. If a line is 100% series compensated it may result into series resonance at
(*a*) power frequency (50 Hz)
(*b*) 100 Hz
(*c*) 150 Hz
(*d*) none of the above.

200. SSR phenomenon is
(*a*) purely electrical
(*b*) purely mechanical
(*c*) purely hydraulic
(*d*) (*a*) and (*b*).

201. The order of the subharmonic during SSR for 50 Hz normal frequency is
(*a*) 25 Hz
(*b*) $16\frac{2}{3}$ Hz
(*c*) 10 Hz
(*d*) none of the above.

202. A transformer rated for 500 KVA 11 KV/0.4 KV has an impedance of 10% and is connected to an infinite bus. The fault level of the transformer is

(*a*) 500 KVA (*b*) 5000 KVA

(*c*) 500 $\sqrt{3}$ KVA (*d*) none of the above.

203. With the help of a reactive compensator it is possible to have

(*a*) constant voltage operation only

(*b*) unity p.f. operation only

(*c*) both constant voltage and unity p.f.

(*d*) either constant voltage or unity p.f.

204. Efficiency of a power transformer under no load condition is approximately

(*a*) 75% (*b*) 50%

(*c*) 25% (*d*) none of the above.

205. A 3-phase induction motor is drawing 5A when a single phasing takes place. The current now drawn by the motor is

(*a*) $5\sqrt{2}$ (*b*) $5\sqrt{3}$

(*c*) $10\sqrt{3}$ (*d*) none of the above.

206. When a d.c. source is switched in a purely inductive circuit the current response is

(*a*) an exponentially rising curve

(*b*) an exponentially decaying curve

(*c*) a straight line passing through the origin

(*d*) a straight line off-set from the origin.

207. The voltage and current in a circuit are given by

$$v = 10 \sin (\omega t - \pi/6)$$
$$i = 10 \sin (\omega t + \pi/6)$$

The power consumed is given as

(*a*) 100 watts (*b*) 50 watts

(*c*) 86.6 watts (*d*) 25 watts

208. In a large interconnected power system, consider three buses having short-circuit capacities 1500 MVA (1), 1200 MVA (2) and 1000 MVA (3) respectively. The voltages of all the buses are 1.0 p.u. If a 3-phase fault takes place on bus 2, the change in bus voltage is described as

(*a*) $\Delta V_1 > \Delta V_2 > \Delta V_3$ (*b*) $\Delta V_1 < \Delta V_3 < \Delta V_2$

(*c*) $\Delta V_1 > \Delta V_3 > \Delta V_2$ (*d*) None of the above.

209. For a long uncompensated line the limit to the line loading is governed by

(*a*) Thermal limit (*b*) Voltage drop

(*c*) Stability limit (*d*) Corona loss.

210. For a lossless line terminated by its surge impedance, the natural reactive power loading is

(*a*) V^2/X (*b*) V^2/Z

(*c*) V^2/Z_c (*d*) Zero.

211. A 60 Hz 320 km lossless line has sending end voltage 1.0 p.u. The receiving end voltage on no load is

(*a*) 1.1 p.u. (*b*) 1.088 p.u.

(*c*) 1.116 p.u. (*d*) none of the above.

212. The impedance of a circuit is given by $z = 3 + j4$, its conductance is given by

(*a*) $\frac{1}{3}$ (*b*) 3/5

(*c*) 3/25 (*d*) 4/5.

213. With 100% series compensation of lines

(*a*) the circuit is series resonant at power frequency

(*b*) low transient voltage

(*c*) high transient current

(*d*) (*a*) and (*c*).

214. The effect of increasing gating angle in a thyristorised controlled reactor is

(*a*) to increase the effective inductance of the reactor

(*b*) to reduce the effective reactive power

(*c*) (*a*) and (*b*)

(*d*) none of the above.

215. The effect of increasing gating angle in a thyristorised controlled reactor is

(*a*) to decrease power loss in thyristor controller

(*b*) to decrease power loss in the reactor

(*c*) to make current wave form less sinusoidal

(*d*) all of the above.

216. The inertia constant H of a synchronous condenser is

(*a*) greater than that of hydro generator

(*b*) greater than that of turbo alternator

(*c*) equal to that of turbo alternator

(*d*) none of the above.

217. The effect of series capacitance compensation is

(*a*) to decrease the virtual surge impedance

(*b*) to decrease the effective length of the line

(*c*) to increase virtual surge impedance loading

(*d*) all of the above.

218. For any fixed degree of series compensation additional capacitive shunt compensation

(*a*) increases the effective length of line

(*b*) increases virtual surge impedance of line

(*c*) decreases virtual surge impedance loading of the line

(*d*) (*b*) and (*c*).

219. For any fixed degree of inductive shunt compensation, additional series capacitive compensation

(*a*) increases the effective length of line

(*b*) increases virtual surge impedance of line

(*c*) decreases virtual surge impedance loading of the line

(*d*) none of the above.

220. With 100% inductive shunt compensation the voltage profile is flat for

(*a*) 100% loading of line (*b*) 50% loading of line

(*c*) Zero loading of line (*d*) None of the above.

221. A loss less line terminated with its surge impedance has

(*a*) Flat voltage profile

(*b*) transmission line angle is greater that actual length of the line

(*c*) transmission line angle is less than the actual length

(*d*) (*a*) and (*b*).

222. The main consideration for higher and higher operating voltage of transmission is to

(*a*) increase efficiency of transmission (*b*) reduce power losses

(*c*) increase power transfer capability (*d*) (*a*) and (*b*).

223. A synchronous generator connected to an infinite bus delivers power at a log p.f. If its excitation is increased

(*a*) the terminal voltage increases (*b*) voltage angle δ increases

(*c*) Current delivered increases (*d*) (*b*) and (*c*).

224. A synchronous motor connected to an infinite bus takes power at a lag p.f. If its excitation is increased

(*a*) the terminal voltage increases (*b*) the load angle increases

(*c*) the p.f. of motor increases (*d*) (*c*) and (*b*).

225. In a pure LC parallel circuit under resonance condition, current drawn from the supply mains is

(*a*) very large (*b*) $V\sqrt{LC}$

(*c*) $V/\sqrt{LC}$ (*d*) Zero.

226. The current at a given point in a certain circuit may be given a function of time as

$$i(t) = -3 + t$$

The total charge passing the point between $t = 99$ sec and $t = 102$ sec is

(*a*) 112 C (*b*) 242.5 C

(*c*) 292.5 C (*d*) 345.6 C.

227. An alternator has a phase sequence of RYB for its phase voltages. In case the field current is reversed, the phase sequence will become

(*a*) RBY (*b*) RYB

(*c*) YRB (*d*) None of the above.

228. An alternator has a phase sequence of RYB for it phase voltage. In case the direction of rotation of alternator is reversed, the phase sequence will become

(*a*) RBY (*b*) RYB

(*c*) YRB (*d*) None of the above.

229. In a circuit the voltage and current are given by $v = (10 + j5)$ and $i = (6 + j4)$. The circuit is

(*a*) inductive (*b*) capacitive

(*c*) resistive (*d*) it could be any of the above.

230. The power in the circuit of problem 229 is

(*a*) 60 watts (*b*) 20 watts

(*c*) 80 watts (*d*) 70 watts.

231. The reactive power in the circuit of problem 229 is

(*a*) 70 VArs (*b*) 60 VArs

(*c*) 10 VArs (*d*) – 10 VArs.

232. A synchronous generator is connected to an infinite bus bar and is initially operating at a lag p.f. If the steam input to the alternator is increased

(*a*) The p.f. of the alternator improves (*b*) reactive power decreases

(*c*) the frequency increase (*d*) (*a*) and (*b*).

233. A synchronous alternator is connected to an infinite bus bar and is initially operating at lead p.f. If the steam input is increased

(*a*) The p.f. increases (*b*) the current delivered increases

(*c*) the frequency increases (*d*) (*a*) and (*c*).

234. A 100 km transmission line is designed for a nominal voltage of 132 kV and consists of one conductor per phase. The line reactance is 0.726 ohm/km. The static transmission capacity of the line, in Megawatts, would be

(*a*) 132 (*b*) 240

(*c*) 416 (*d*) 720.

235. The per unit impedance of a circuit element is 0.15. If the base kV and base MVA are halved, then the new value of the per unit impedance of the circuit element will be

(*a*) 0.075 (*b*) 0.15

(*c*) 0.30 (*d*) 0.600.

236. If the effect of earth is taken into account, then the capacitance of line to ground

(*a*) decreases (*b*) increases

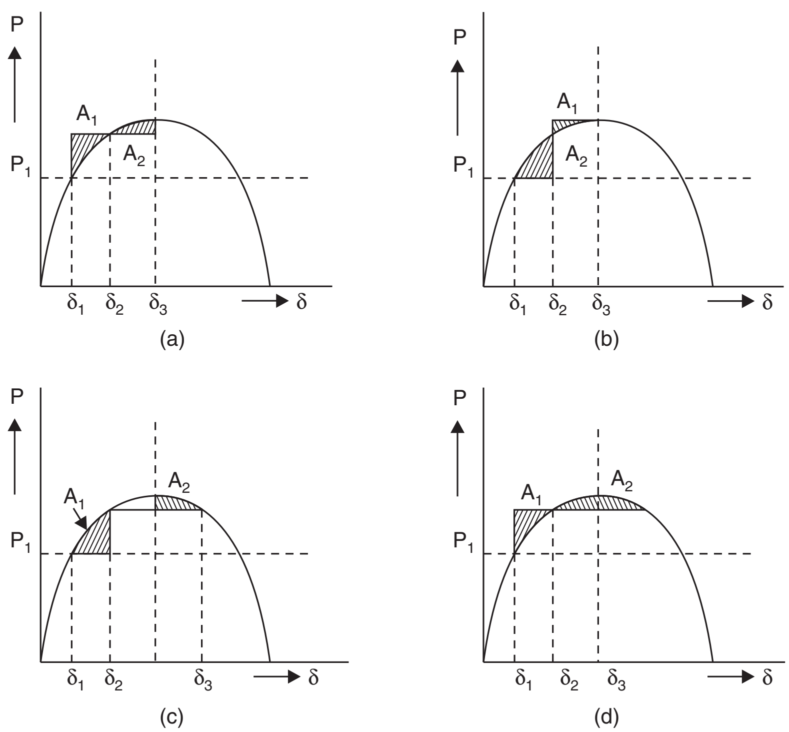
(*c*) remains unaltered (*d*) becomes infinite.

237. A single line to ground fault occurs on an unloaded generator in phase a. If $x_d = x_2 = 0.25$ p.u., $x_0 = 0.15$ p.u., reactance connected in the neutral, $x_n = 0.05$ p.u. and the initial prefault voltage is 1.0 p.u., then the magnitude of the fault current will be

(*a*) 3.75 p.u. (*b*) 1.54 p.u.

(*c*) 1.43 p.u. (*d*) 1.25 p.u.

238. Principle of Equal-area Criterion is to be applied to determine, for a given initial load P_1, the maximum amount of sudden increase in load ΔP, to maintain transient stability of a cylindrical rotor synchronous motor operating from an infinite bus. Applying this criterion (in each case the area A_1 = area A_2). Which one of the following diagrams is correct ?



239. A thyrite type lightning arrestor

(*a*) blocks the surge voltage appearing in a line

(*b*) absorbs the surge voltage appearing in a line

(*c*) offers a low resistance path to the surge appearing in line

(*d*) returns the surge back to the source.

240. A 66-kV system has string insulator having five discs and the earth to disc capacitance ratio of 0.10. The string efficiency will be

(*a*) 89% (*b*) 75%

(*c*) 67% (*d*) 55%.

241. Which of the following statements regarding corona are true ?

1. It causes radio interference
2. It attenuates lightning surges
3. It amplifies switching surges
4. It causes power loss
5. It is more prevalent in the middle conductor of a transmission line employing a flat conductor configuration. Select the correct answer using the codes given below:

(*a*) 1, 3 and 5 (*b*) 2, 3 and 4

(*c*) 1, 2, 4 and 5 (*d*) 2, 3, 4 and 5.

242. ϕ_{1m}, ϕ_{2m} = the fluxes produced by the two portions of the shaded pole,

θ = the angle between ϕ_{1m} and ϕ_{2m}

R = resistance of the disc

The torque produced in an induction relay would be proportional to which of the following ?

1. ϕ_{1m} and ϕ_{2m}
2. $1/R$
3. R
4. $\sin \theta$

Select the correct answer using the codes given below:

(*a*) 1, 2 and 4 (*b*) 1, 3 and 4

(*c*) 1 and 2 (*d*) 2 and 4.

243. Match list 1 with list 2 and select the correct answer using the codes given below the lists:

List 1

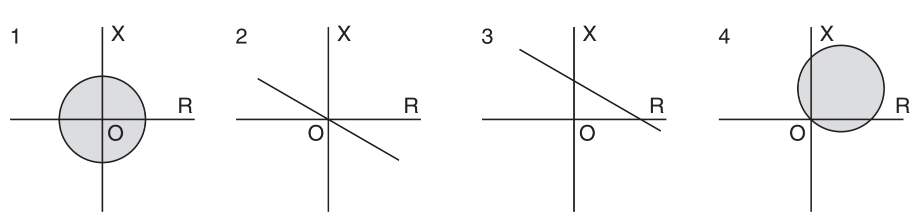
A. Mho relay

B. Plain impedance relay

C. Directional relay

D. Angle impedance relay

List 2



Codes:

(*a*)	*A*	*B*	*C*	*D*
	4	3	2	1
(*b*)	*A*	*B*	*C*	*D*
	4	1	2	3
(*c*)	*A*	*B*	*C*	*D*
	3	2	1	4
(*d*)	*A*	*B*	*C*	*D*
	3	2	4	1

244. For a *Y*-delta transformer with *Y*-side grounded, the zero sequence current

(*a*) has no path to ground

(*b*) exists in the lines on the delta side

(*c*) exists in the lines on the *Y* side

(*d*) exists in the lines on both *Y* and delta sides.

245. When there is a change in load in a power station having a number of generator units operating in parallel, the system frequency is controlled by

(*a*) adjusting the steam input to the units

(*b*) adjusting the field-excitation of the generators

(*c*) changing the load divisions between the units

(*d*) injecting reactive power at the station bus bar.

246. A 2 KVA transformer has iron loss of 150 Watts and full-load copper loss of 250 Watts. The maximum efficiency of the transformer would occur when the total loss is

(*a*) 500 W (*b*) 400 W

(*c*) 300 W (*d*) 275 W.

247. If the frequency of input voltage of a transformer is increased keeping the magnitude of voltage unchanged, then

(*a*) both hysteresis loss and eddy current loss in the core will increase

(*b*) hysteresis loss will increase but eddy current loss will decrease

(*c*) hysteresis loss will decrease but eddy current loss will increase

(*d*) hysteresis loss will decrease but eddy current loss will remain unchanged.

248. For successful parallel operation of two single-phase transformers, the most essential condition is that their

(*a*) percentage impedances are equal

(*b*) polarities are properly connected

(*c*) turns ratios are exactly equal

(*d*) KVA ratings are equal.

249. The most appropriate operating speeds in rpm of generators used in Thermal, Nuclear and Hydro-power plants would respectively be

(*a*) 3000, 300 and 1500 (*b*) 3000, 3000 and 300

(*c*) 1500, 1500 and 3000 (*d*) 1000, 900 and 750.

250. The operation of a nuclear reactor is controlled by controlling the multiplication-factor (K), defined as

$$K = \frac{\text{number of neutrons of any one Generation}}{\text{number of neutrons of immediately preceding generation}}$$

The power-level of the reactor can be increased by

(*a*) raising the value of K above 1 and, keeping it at that raised value

(*b*) raising the value of K above 1, but later bringing it back to $K = 1$

(*c*) lowering the value of K below 1 and, keeping it at that lowered value

(*d*) lowering the value of K below 1, but later bringing it back to $K = 1$.

251. The inductance of single-phase two-wire power transmission line per kilometer gets doubled when the

(*a*) distance between the wires is doubled

(*b*) distance between the wires is increased four fold.

(*c*) distance between the wires is increased as square of original distance

(*d*) radius of the wire is doubled.

252. The flow-duration curve at a given head of a hydro-electric plant is used to determine the

(*a*) total power available at the site

(*b*) total units of energy available

(*c*) load-factor at the plant

(*d*) diversity-factor for the plant.

253. Match List I with List II and select the correct answer using the codes given below the lists:

List I	*List II*
A. Thyrite arrester	1. Tower location
B. Sag template	2. Cross bonding
C. Cable sheaths	3. Restriking voltage
D. Circuit breaker	4. Non-linear resistor.

Codes:

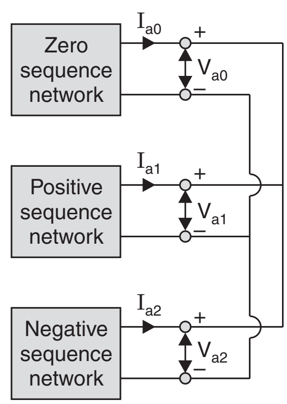
	A	*B*	*C*	*D*
(*a*)	4	1	3	2
(*b*)	4	1	2	3

(*c*)	*A*	*B*	*C*	*D*
	1	4	3	2
(*d*)	*A*	*B*	*C*	*D*
	4	3	1	2

254. The connection diagram of sequence networks for a particular fault on a power system network is given in the figure. The type of the fault is

(*a*) single line to ground fault (*b*) double line to ground fault

(*c*) line to line fault (*d*) open circuit.



255. The inertia constant of a 100 MVA, 11 kV water-wheel generator is 4. The energy stored in the rotor at the synchronous speed is

(*a*) 400 MJ (*b*) 400 kJ

(*c*) 25 MJ (*d*) 25 kJ.

256. A large ac generator, supplying power to an infinite bus, has a sudden short-circuit occuring at its terminals. Assuming the prime mover input and the voltage behind the transient reactance to remain constant immediately after the fault, the acceleration of the generator rotor is

(*a*) inversely proportional to the moment of inertia of the machine

(*b*) inversely proportional to the square of the voltage

(*c*) directly proportional to the square of the short-circuit current

(*d*) directly proportional to the short-circuit power.

257. Match List I (Devices) with List II (Application) and select the correct answer using the codes given below the lists:

List I	*List II*
A. Microprocessor	1. Monitoring surge voltage
B. Breather	2. Digital protection
C. Magnetic links	3. Monitoring surge current
D. Klydonograph	4. Power transformer

Codes:

(*a*)	*A*	*B*	*C*	*D*
	4	2	1	3
(*b*)	*A*	*B*	*C*	*D*
	4	2	3	1
(*c*)	*A*	*B*	*C*	*D*
	2	4	1	3
(*d*)	*A*	*B*	*C*	*D*
	2	4	3	1

258. A lightning arrestor connected between the line and earth in a power system

(*a*) protects the terminal equipment against travelling surges

(*b*) protects the transmission line against lightning stroke

(*c*) suppresses high frequency oscillations in the line

(*d*) reflects back the travelling wave approaching it.

259. Corona loss can be reduced by the use of hollow conductor, because

(*a*) the current density is reduced

(*b*) the eddy-current in the conductor is eliminated

(*c*) for a given cross-section, the radius of the conductor is increased

(*d*) of better ventilation in the conductor.

260. The insulation of modern EHV lines is designed based on

(*a*) the lightning voltage (*b*) corona

(*c*) radio interference (*d*) switching voltage.

261. While using air-blast circuit breaker, current chopping is a phenomenon often observed when

(*a*) a long overhead line is switched off

(*b*) a bank of capacitors is switched off

(*c*) a transformer on no-load is switched off

(*d*) a heavy load is switched off.

262. The most efficient torque-producing actuating structure for induction-type relay is

(*a*) shaded-pole structure (*b*) watt-hour-meter structure

(*c*) induction-cup structure (*d*) single-induction-loop structure.

263. A power system network with a capacity of 100 MVA has a source impedance of 10% at a point. The fault level at that point is

(*a*) 10 MVA (*b*) 30 MVA

(*c*) 3000 MVA (*d*) 1000 MVA.

264. In a power station, the cost of generation of power reduces most effectively when

(*a*) diversity factor alone increases

(*b*) both diversity factor and load factor increase

(*c*) load factor alone increases

(*d*) both diversity factor and load factor decrease.

265. In the optimum generator scheduling of different power plants, the minimum fuel cost is obtained when

(*a*) only the incremental fuel cost of each plant is the same

(*b*) the penalty factor of each plant is the same

(*c*) the ratio of the incremental fuel cost to the penalty factor of each plant is the same

(*d*) the incremental fuel cost of each plant multiplied by its penalty factor is the same.

266. Shunt compensation in an EHV line is used to

(*a*) improve stability

(*b*) reduce fault level

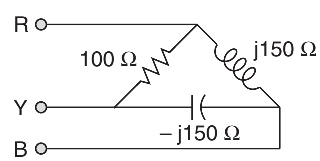
(*c*) improve the voltage profile

(*d*) substitute for synchronous phase modifier.

267. A set of 3 equal resistors, each of value R_X, connected in star across RYB in place of load as shown in the given figure consumes the same power as the unbalanced delta connected load. The value of R_X is

(*a*) 33.33 Ω (*b*) 100 Ω

(*c*) 173.2 Ω (*d*) 300 Ω.



268. In a 400 kV power network, 360 kV is recorded at 400 kV bus. The reactive power absorbed by a shunt reactor rated for 50 MVAR, 400 kV connected at the bus is

(*a*) 61.73 MVAR (*b*) 55.56 MVAR

(*c*) 45.0 MVAR (*d*) 40.5 MVAR.

269. Which one of the following statements is true ?

(*a*) Steady-state stability limit is greater than transient stability limit

(*b*) Steady-state stability limit is equal to transient stability limit

(*c*) Steady-state stability limit is less than the transient stability limit

(*d*) No generalisation can be made regarding the equality or otherwise of the steady-state stability limit and transient stability limit.

270. Lightning arresters are used in power systems to protect electrical equipments against

(*a*) direct strokes of lightning

(*b*) power frequency overvoltages

(*c*) overvoltages due to indirect lightning stroke

(*d*) overcurrents due to lightning stroke.

271. The voltages across the various discs of a string of suspension insulators having identical discs is different due to

(*a*) surface leakage currents (*b*) series capacitance

(*c*) shunt capacitances to ground (*d*) series and shunt capacitances.

272. Compared with a solid conductor of the same radius, corona appears on a stranded conductor at a lower voltage, because stranding

(*a*) assists ionisation

(*b*) makes the current flow spirally about the axis of the conductor

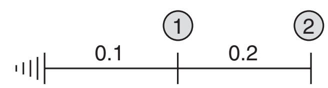
(*c*) produces oblique sections to a plane perpendicular to the axis of the conductor

(*d*) produces surfaces of smaller radius.

273. The bus admittance matrix of the network shown in the given figure, for which the marked parameters are per unit impedance, is

(*a*) $\begin{bmatrix} 0.3 & -0.2 \\ -0.2 & 0.2 \end{bmatrix}^{-1}$ (*b*) $\begin{bmatrix} 0.3 & 0.2 \\ 0.2 & 0.2 \end{bmatrix}$

(*c*) $\begin{bmatrix} 0.3 & -0.2 \\ -0.2 & 0.2 \end{bmatrix}$ (*d*) $\begin{bmatrix} 15 & -5 \\ -5 & 5 \end{bmatrix}$.



274. Buses for load flow studies are classified as (*i*) the load bus, (*ii*) the generator bus and (*iii*) the slack bus

The correct combination of the pair of quantities specified having their usual meaning for different buses is

	Load bus	Generator bus	Slack bus
(*a*)	$P, \lvert V \rvert$	P, Q	P, δ
(*b*)	P, Q	$P, \lvert V \rvert$	$\lvert V \rvert, \delta$
(*c*)	$\lvert V \rvert, Q$	P, δ	P, Q
(*d*)	P, δ	$Q, \lvert V \rvert$	Q, δ

275. A Bucholz relay is used for

(*a*) protection of a transformer against all internal faults

(*b*) protection of a transformer against all external faults

(*c*) protection of a transformer against both internal and external faults

(*d*) protection of induction motors.

276. The line trap unit employed in carrier current relaying

(*a*) offers high impedance to 50 Hz power frequency signals

(*b*) offers high impedance to carrier frequency signals

(*c*) offers low impedance to carrier frequency signals

(*d*) offers low impedance to carrier frequency signals.

277. Match List I with List II and select the correct answer using the codes given below the lists:

List I	*List II*
(Equipment)	*(Function)*
A. Circuit Breaker	1. Voltage control
B. Lightning Arrester	2. Power Control
C. Governor	3. Overvoltage protection
D. Exciter	4. Overcurrent protection.

Codes :

(*a*)	*A*	*B*	*C*	*D*
	1	2	3	4
(*b*)	*A*	*B*	*C*	*D*
	4	1	2	3
(*c*)	*A*	*B*	*C*	*D*
	2	3	4	1
(*d*)	*A*	*B*	*C*	*D*
	4	3	2	1

278. To prevent maloperation of differentially connected relay while energising a transformer, the relay restraining coil is biased with

(*a*) second harmonic current (*b*) third harmonic current

(*c*) fifth harmonic current (*d*) seventh harmonic current.

279. The incremental fuel cost for two generating units are given by

$IC_1 = 25 + 0.2\ PG_1$

$IC_2 = 32 + 0.2\ PG_2$, where PG_1 and PG_2 are real power generated by the units.

The economic allocation for a total load of 250 MW, neglecting transmission loss is given by

(*a*) $PG_1 = 140.25$ MW, $PG_2 = 109.75$ MW

(*b*) $PG_1 = 109.75$ MW, $PG_2 = 140.25$ MW

(*c*) $PG_1 = PG_2 = 125$ MW

(*d*) $PG_1 = 100$ MW, $PG_2 = 150$ MW.

280. The synchronisation coefficient between two area of a 2-area power system is (symbols have usual meanings)

(*a*) $\dfrac{\partial P}{\partial |V|}$ (*b*) $\dfrac{\partial P}{\partial \delta}$

(*c*) $\dfrac{\partial P}{\partial f}$ (*d*) $\dfrac{\partial P}{\partial Q}$.

281. In a high voltage dc transmission scheme, reactive power is needed both for the rectifier at the sending-end and, for the inverter at the reciving-end. During the operation of such a dc link the rectifier receives

(*a*) lagging reactive power and the inverter supplies leading reactive power

(*b*) leading reactive power and the inverter supplies lagging reactive power

(*c*) lagging reactive power and the inverter supplies lagging reactive power

(*d*) leading reactive power and the inverter supplies leading reactive power.

282. Transmission lines are transposed to

(*a*) reduce copper loss

(*b*) reduce skin effect

(*c*) prevent interference with neighbouring telephone lines

(*d*) prevent short-circuit between any two lines.

283. A single-phase transmission line of impedance j 0.8 ohm supplies a resistive load of 500 A at 300 V. The sending-end power factor is

(*a*) unity (*b*) 0.8 lagging

(*c*) 0.8 leading (*d*) 0.6 lagging.

284. The impedance value of a generator is 0.2 pu on a base value of 11 KV, 50 MVA. The impedance value for a base value of 22 KV, 150 MVA is

(*a*) 0.15 pu (*b*) 0.2 pu

(*c*) 0.3 pu (*d*) 2.4 pu.

285. For a transmission line with resistance, R, reactance X, and negligible capacitance, the transmission constant A is

(*a*) 0 (*b*) 1

(*c*) $R + jx$ (*d*) $R + X$.

286. Four identical alternators each rated for 20 MVA, 11 kV having a subtransient reactance of 16% are working in parallel. The short- circuit level at the bus-bars is

(*a*) 500 MVA (*b*) 400 MVA

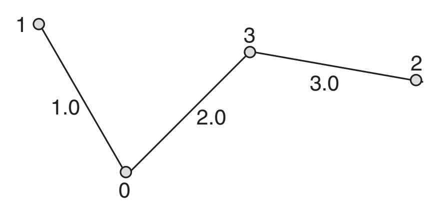
(*c*) 125 MVA (*d*) 80 MVA.

287. Consider the network shown in the following figure:

The bus numbers and impedances are marked

The bus impedance matrix of this network is

(*a*) $\begin{array}{c|c} & \begin{array}{ccc} 1 & 2 & 3 \end{array} \\ \begin{array}{c} 1 \\ 2 \\ 3 \end{array} & \begin{bmatrix} 1.0 & 0 & 0 \\ 0 & 2.0 & 0 \\ 0 & 0 & 5.0 \end{bmatrix} \end{array}$



(*b*) $\begin{array}{c|c} & \begin{array}{ccc} 1 & 2 & 3 \end{array} \\ \begin{array}{c} 1 \\ 2 \\ 3 \end{array} & \begin{bmatrix} 1.0 & 0 & 0 \\ 0 & 2.0 & 0 \\ 0 & 0 & 3.0 \end{bmatrix} \end{array}$

(c) $\begin{array}{cc} & \begin{array}{ccc} 1 & 2 & 3 \end{array} \\ \begin{array}{c} 1 \\ 2 \\ 3 \end{array} & \begin{bmatrix} 1.0 & 0.0 & 0.0 \\ 0.0 & 5.0 & 2.0 \\ 0.0 & 2.0 & 2.0 \end{bmatrix} \end{array}$

(d) $\begin{array}{cc} & \begin{array}{ccc} 1 & 2 & 3 \end{array} \\ \begin{array}{c} 1 \\ 2 \\ 3 \end{array} & \begin{bmatrix} 1.0 & 0.0 & 0.0 \\ 0.0 & 2.0 & 2.0 \\ 0.0 & 2.0 & 5.0 \end{bmatrix} \end{array}$.

288. In a power system with negligible resistance, the fault current at a point is 8.00 pu. The series reactance to be included at the fault point to limit the short-circuit current to 5.00 pu is

(a) 3.000 pu (b) 0.200 pu
(c) 0.125 pu (d) 0.075 pu.

289. The 'equal area criterion' for the determination of transient stability of a synchronous machine connected to an infinite bus

(a) ignores line as well as synchronous machine resistances
(b) assumes accelerating power acting on the rotor as constant
(c) ignores the effect of voltage regulator and governor but considers the inherent damping present in the machine
(d) takes into consideration the possibility of machine losing synchronisation after it has survived during the first swing.

290. A voltage of 1000 kV is applied to an overhead line with its receiving-end open. If the surge impedance of the line is 500 ohms, then the total surge power in the line is

(a) 2000 MW (b) 500 MW
(c) 2 MW (d) 0.5 MW.

291. A short transmission line, having its line-impedance angle as θ, is delivering a given power at the receiving-end at a lagging power-factor angle of ϕ. Which one of the following is a set of conditions for which this line will have maximum- and zero-regulation ?

Maximum Regulation	Zero Regulation
(a) $\phi = \theta$	$\phi - \theta = \pi/2$
(b) $\phi - \theta = \pi/2$	$\phi = \theta$
(c) $\phi = \theta$	$\phi + \theta = \pi/2$
(d) $\phi + \theta = \pi/2$	$\phi = \theta$.

292. Impulse ratios of insulators and lightning arresters should be

(a) both low (b) high and low respectively
(c) low and high respectively (d) both high.

293. Bundled conductors in EHV transmission system provide

(a) reduced capacitance (b) increased capacitance
(c) increased inductance (d) increased voltage gradient.

294. The charging current in a transmission line increases due to corona effect because corona increases

(*a*) line current (*b*) effective line voltage

(*c*) power loss in lines (*d*) the effective conductor diameter.

295. For a 15-bus power system with 3 voltage controlled bus, the size of Jacobian matrix is

(*a*) 11 × 11 (*b*) 12 × 12

(*c*) 24 × 24 (*d*) 28 × 28.

296. If the fault current is 2 kA, the relay setting is 50% and the C.T. ratio is 400/5, then the plug setting multiplier of a relay will be

(*a*) 5 (*b*) 7

(*c*) 8 (*d*) 10.

297. The bus-bars of each of the two alternators of 15% reactance each, are interconnected through tie-bar reactors of 15% each. The equivalent impedance to fault current for a 3-phase fault in any alternator bus-bar will be

(*a*) 75% (*b*) 10%

(*c*) 11.25% (*d*) 15%.

298. A power plant has a maximum demand of 15 MW. The load factor is 50% and the plant factor is 40%. The operating reserve is

(*a*) 3.0 MW (*b*) 3.75 MW

(*c*) 6.0 MW (*d*) 7.5 MW.

299. The incremental cost characteristic of the two units in a plant are

$$IC_1 = 0.1\ P_1 + 8.0 \text{ Rs./MWh}$$

$$IC_2 = 0.15\ P_2 + 3.0 \text{ Rs./MWh}$$

When the total load is 100 MW, the optimum sharing of the load is

	P_1	P_2
(*a*)	40 MW	60 MW
(*b*)	33.3 MW	66.7 MW
(*c*)	60 MW	40 MW
(*d*)	66.7 MW	33.3 MW.

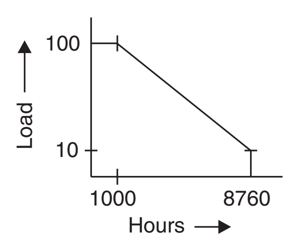
300. The load curve of a system is shown in the figure. The load factor of the system is:

(*a*) 1.66%

(*b*) 6.013%

(*c*) 16.6%

(*d*) 60.13%.



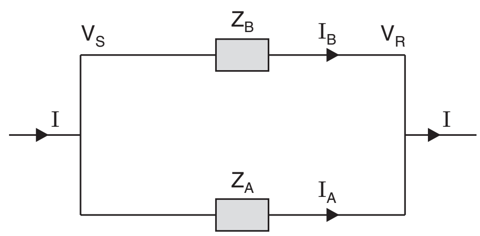
301. Control rods used in nuclear reactors are made of

(*a*) zirconium (*b*) boron

(*c*) beryllium (*d*) lead.

302. Consider two parallel short transmission lines of impedances Z_A and Z_B respectively as shown in the figure. Currents I_A and I_B are both lagging and the sending-end voltage is V_s. If the reactance to resistance ratio of both impedances Z_A abd Z_B are equal, then the total current 'I' will

(a) lag both I_A and I_B
(b) lead both I_A and I_B
(c) lag one of I_A and I_B but lead the other
(d) be in phase with both I_A and I_B.



303. In a 3-phase extra-high voltage cable, a metallic screen around each core-insulation is provided to

(a) facilitate heat dissipation
(b) give mechanical strength
(c) obtain radial electric stress
(d) obtain longitudinal electric stress.

304. Galloping in transmission line conductors arises generally due to

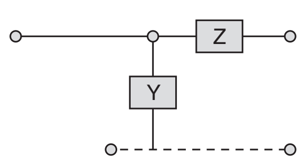
(a) asymmetrical layers of ice formation
(b) vortex phenomenon in light winds
(c) heavy weight of the line conductors
(d) adoption of horizontal conductor configurations.

305. In a matrix form, the equation of a 4-terminal network representing a transmission line is given by

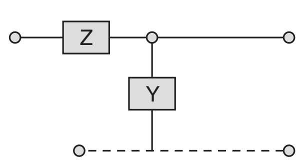
$$\begin{bmatrix} V_s \\ I_s \end{bmatrix} = \begin{bmatrix} A & B \\ C & D \end{bmatrix} \begin{bmatrix} V_R \\ I_R \end{bmatrix}$$

The two networks considered are

(A)



(B)



The plausible transfer matrix for the networks (A) and (B) could be:

(*i*) $\begin{bmatrix} 1 & YZ \\ 0 & Z \end{bmatrix}$ (*ii*) $\begin{bmatrix} Y & 0 \\ YZ & 1 \end{bmatrix}$

(*iii*) $\begin{bmatrix} 1 & Z \\ Y & 1+YZ \end{bmatrix}$ (*iv*) $\begin{bmatrix} 1+YZ & Z \\ Y & 1 \end{bmatrix}$.

The correct combination for the two networks (*A*) and (*B*), would be:

(*a*) (*i*) and (*ii*) (*b*) (*i*) and (*iii*)

(*c*) (*ii*) and (*iv*) (*d*) (*iii*) and (*iv*).

306. The incremental generating costs of two generating units are given by

$$IC_1 = 0.1X + 20 \text{ Rs./MWhr}$$

$$IC_2 = 0.15Y + 18 \text{ Rs./MWhr.}$$

where X and Y are power (in MW) generated by the two units. For a total demand of 300 MW, the values (in MW) of X and Y will be respectively

(*a*) 172 and 128 (*b*) 128 and 172

(*c*) 175 and 125 (*d*) 200 and 100.

307. Consider the following statements:

To provide reliable protection for a distribution transformer against overvoltages using lightning arresters, it is essential that the

1. lead resistance is high
2. distance between the transformer and the arrester is small
3. transformer and the arrester have a common inter-connecting ground
4. spark overvoltage of the arrester is greater than the residual voltage.

Of these statements

(*a*) 1, 3 and 4 are correct (*b*) 2 and 3 are correct

(*c*) 2, 3 and 4 are correct (*d*) 1 and 4 are correct.

308. The reflection coefficient of a short-circuited line for voltage is

(*a*) – 1 (*b*) + 1

(*c*) 0.5 (*d*) zero.

309. The propagation constant of a transmission line is:

$$0.15 \times 10^{-3} + j1.5 \times 10^{-3}$$

The wavelength of the travelling wave is

(*a*) $\dfrac{15 \times 10^{-3}}{2\pi}$ (*b*) $\dfrac{2\pi}{15 \times 10^{-3}}$

(*c*) $\dfrac{15 \times 10^{-3}}{\pi}$ (*d*) $\dfrac{\pi}{15 \times 10^{-3}}$.

310. Hollow conductors are used in transmission lines to

(*a*) reduce weight of copper (*b*) improve stability

(*c*) reduce corona (*d*) increase power transmission capacity.

311. In the solution of load-flow equation, Newton-Raphon (NR) method is superior to the Gauss-Seidal (GS) method, because the

(*a*) time taken to perform one iteration in the NR method is less when compared to the time taken in the GS method

(*b*) number of iteration required in the NR method is more when compared to that in the GS method

(*c*) number of iterations required is not independent of the size of the system in the NR method

(*d*) convergence characteristic of the NR method are not affected by the selection of slack bus.

312. A power system network consists of three elements 0-1, 1-2 and 2-0 of per unit impedances 0.2, 0.4 and 0.4 respectively. It's bus impedance matrix is given by

(*a*) $\begin{array}{cc} & \begin{array}{cc} 1 & 2 \end{array} \\ \begin{array}{c} 1 \\ 2 \end{array} & \begin{bmatrix} 7.5 & -2.5 \\ -2.5 & 5.0 \end{bmatrix} \end{array}$ (*b*) $\begin{array}{cc} & \begin{array}{cc} 1 & 2 \end{array} \\ \begin{array}{c} 1 \\ 2 \end{array} & \begin{bmatrix} 0.16 & 0.08 \\ 0.08 & 0.24 \end{bmatrix} \end{array}$

(*c*) $\begin{array}{cc} & \begin{array}{cc} 1 & 2 \end{array} \\ \begin{array}{c} 1 \\ 2 \end{array} & \begin{bmatrix} 0.16 & -0.08 \\ -0.08 & 0.24 \end{bmatrix} \end{array}$ (*d*) $\begin{array}{cc} & \begin{array}{cc} 1 & 2 \end{array} \\ \begin{array}{c} 1 \\ 2 \end{array} & \begin{bmatrix} 0.6 & 0.4 \\ 0.4 & 0.8 \end{bmatrix} \end{array}$.

313. Zero sequence currents can flow from a line into a transformer bank if the windings are in

(*a*) grounded star/delta (*b*) delta/star

(*c*) star/grounded star (*d*) delta/delta.

314. When a line-to-ground fault occurs, the current in a faulted phase is 100 *A*. The zero sequence current in this case will be

(*a*) zero (*b*) 33.3 *A*

(*c*) 66.6 *A* (*d*) 100 *A*.

315. When a 50 MVA, 11 kV, 3-phase generator is subjected to a 3-phase fault, the fault current is $-j5$ pu (per unit). When it is subjected to a line-to-line fault, the positive sequence current is $-j4$ pu. The positive and negative sequence reactances are respectively

(*a*) $j0.2$ and $j0.05$ pu (*b*) $j0.2$ and $j0.25$ pu

(*c*) $j0.25$ and $j0.25$ pu (*d*) $j0.05$ and $j0.05$ pu.

316. The power generated by two plants are:

$P_1 = 50$ MW, $P_2 = 40$ MW. If the loss coefficients are $B_{11} = 0.001$, $B_{22} = 0.0025$ and $B_{12} = -0.0005$, then the power loss will be

(*a*) 5.5 MW (*b*) 6.5 MW

(*c*) 4.5 MW (*d*) 8.5 MW.

317. The following data pertain to two alternators working in parallel and supplying a total load of 80 MW:

Machine 1 : 40 MVA with 5% speed regulation

Machine 2 : 60 MVA with 5% speed regulation

The load sharing between machines 1 and 2 will be:

(*a*) $\frac{P_1}{48 \text{ MW}}, \frac{P_2}{32 \text{ MW}}$ (*b*) 40 MW, 40 MW

(*c*) 30 MW, 50 MW (*d*) 32 MW, 48 MW.

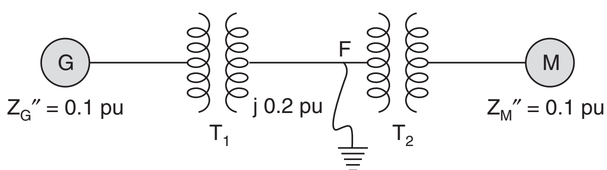
318. The per unit impedance of a synchronous machine is 0.242. If the base voltage is increased by 1.1 times, the per unit value will be

(*a*) 0.266 (*b*) 0.242

(*c*) 0.220 (*d*) 0.200

319. The following figure shows the single line diagram of a power system with all reactances marked in per unit (pu) on the same base:

The system is on no-load when a 3-phase fault occurs at 'F' on the high voltage side of the transformer T_2. The fault current will be



(*a*) $-j0.8187$ pu (*b*) $+j0.8187$ pu

(*c*) $-j8.1871$ pu (*d*) $+j8.1871$ pu.

320. Rated breaking capacity (MVA) of a circuit breaker is equal to

(*a*) the product of rated breaking current (kA) and rated voltage (kV)

(*b*) the product of rated symmetrical breaking current (kA) and rated voltage (kV)

(*c*) the product of breaking current (kA) and fault voltage (kV)

(*d*) twice the value of rated current (kA) and rated voltage (kV).

321. If, in a short transmission line, resistance and inductive reactance are found to be equal and regulation appears to be zero, then the load will

(*a*) have unity power factor (*b*) have zero power factor

(*c*) be 0.707 leading (*d*) be 0.707 lagging.

322. A hydel power plant of run-off-river type should be provided with a pondage so that the

(*a*) firm-capacity of the plant is increased

(*b*) operating head is controlled

(*c*) pressure inside the turbine casing remains constant

(*d*) kinetic energy of the running water is fully utilised.

323. If within an untransposed 3-phase circuit of a transmission line, the series impedance of each of the conductor is considered, it is found to contain resistive terms of the form

$K \log_e \left(\frac{d_{12}}{d_{13}}\right)$, K being a constant and d_{12} and d_{13} etc., being spacings between the conductors. These terms represent power transfer from one phase to another. The sum of these term over the three phases is

(*a*) three times the average (*b*) $\sqrt{3}$ times the average

(*c*) one-third of the average (*d*) zero.

324. Match List I (Parameter) with List II (Effect) and select the correct answer using the codes given below the Lists:

List I	*List II*
A. Percent power lost in transmission	1. Decreases with system voltage
B. For a given current density the conductor size	2. Reduces with line length
C. Power handling capacity of a line at a given voltage	3. Remains independent of line length
D. Surge impedance of a transmission line	4. Increases with line length

Code:

	A	B	C	D
(*a*)	1	2	4	3
(*b*)	3	4	2	1
(*c*)	3	2	4	1
(*d*)	1	4	2	3

325. "Expanded ACSR" are conductor composed of

(*a*) larger diameter individual strands for a given cross-section of the aluminium strands

(*b*) larger diameter of the central steel strands for a given overall diameter of the conductor

(*c*) larger diameter of the aluminium strand only for a given overall diameter of the conductor

(*d*) a filler between the inner steel and the outer aluminium strands to increase the overall diameter of the conductor.

326. Which of the following are the advantages of interconnected operation of power systems ?

1. Less reserve capacity requirement
2. More reliability
3. High power factor
4. Reduction in short-circuit level.

Select the correct answer using the codes given below:

Codes:

(*a*) 1 and 2 (*b*) 2 and 3

(*c*) 3 and 4 (*d*) 1 and 4.

327. Steady-state stability of a power system is improved by

(*a*) reducing fault clearing time

(*b*) using double circuit line instead of single circuit line

(*c*) single pole switching

(*d*) decreasing generator inertia.

328. Equal area criterion gives the information regarding

(*a*) stability region (*b*) absolute stability

(*c*) relative stability (*d*) swing curves.

329. With a number of generators of MVA capacities $S_1, S_2, ..., S_n$ and inertia constants $H_1, H_2, ..., H_n$ respectively connected to the same bus bar in a station, the inertia constant of the equivalent machine on a base of S_b is given by

(*a*) $\sum_{i=1}^{n} H_i \, S_i$ (*b*) $\sum_{i=1}^{n} \frac{S_b}{S_i} . H_i$

(*c*) $\sum_{i=1}^{n} \frac{S_i}{S_b} . H_i$ (*d*) $\sum_{i=1}^{n} \frac{S_i}{S_b} . \frac{1}{H_i}$.

330. The critical clearing time of a fault in power systems is related to

(*a*) reactive power limit (*b*) short-circuit current limit

(*c*) steady-state stability limit (*d*) transient stability limit.

331. The use of fast acting relays and circuit breakers for clearing a sudden short-circuit on a transmission link between a generator and receiving-end bus improves the transient stability of the machine because

(*a*) short-circuit current becomes zero

(*b*) post-fault transfer impedance attains a value higher than that during the fault

(*c*) ordinate of the post-fault power-angle characteristic is higher than that of during-fault characteristic

(*d*) voltage behind the transient reactance increases to a higher value.

332. Consider the following statements:

The transient stability of the power system under unbalanced fault conditions can be effectively improved by

1. Excitation control
2. phase-shifting transformer
3. single-pole switching of ciruit breakers
4. increasing the turbine input.

Of these statements

(*a*) 1 and 2 are correct
(*b*) 2 and 3 are correct
(*c*) 3 and 4 are correct
(*d*) 1 and 3 are correct.

333. The non-uniform distribution of voltage across the units in a string of suspension type insulators is due to

(*a*) unequal self-capacitance of the units
(*b*) non-uniform distance of separation of the units from the tower body
(*c*) the existance of stray capacitance between the metallic junctions of the units and the tower body
(*d*) non-uniform distance between the cross-arm and the units.

334. Whenever the conductors are dead-ended or there is a change in the direction of transmission line, the insulators used are of the

(*a*) pin type
(*b*) suspension type
(*c*) strain type
(*d*) shackle type.

335. Consider the following statements:

In the case of suspension-type insulators, the string efficiency can be improved by

1. using a longer cross arm
2. using a guard ring
3. grading the insulator discs
4. reducing the cross-arm length.

Of these statements

(*a*) 1, 2 and 3 are correct
(*b*) 2, 3 and 4 are correct
(*c*) 2 and 4 are correct
(*d*) 1 and 3 are correct.

336. In a power system, each bus or node is associated with four quantities, namely

1. real power
2. reactive power
3. bus-voltage magnitude and
4. phase-angle of the bus voltage.

For load-flow solution, among these four, the number of quantities to be specified is

(*a*) any one
(*b*) any two
(*c*) any three
(*d*) all the four.

337. Match List I (Type of relay) with List II (For the protection of) and select the correct answer using the codes given below in the Lists:

List I	*List II*
A. Buchholtz relay	1. Feeder
B. Translay relay	2. Transformer
C. Negative sequence relay	3. Generator
D. Directional over current relay	4. Long Overhead line

Codes:

(*a*)	*A*	*B*	*C*	*D*
	1	2	3	4
(*b*)	*A*	*B*	*C*	*D*
	2	1	3	4
(*c*)	*A*	*B*	*C*	*D*
	2	1	4	3
(*d*)	*A*	*B*	*C*	*D*
	1	4	2	3

338. A line trap in carrier current relaying tuned to carrier frequency presents

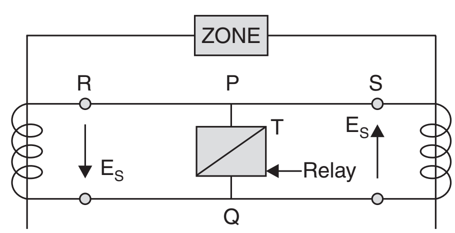
(*a*) high impedance to carrier frequency but low impedance to power frequency

(*b*) low impedance to both carrier and power frequency

(*c*) high impedance to both carrier and power frequency

(*d*) low impedance to carrier frequency but high impedance to power frequency.

339. The basic circuit of circulating current system of protection is shown in the figure. To improve the through-fault stability, a stabilizing resistor is connected between the points



(*a*) *R* and *P* in series (*b*) *P* and *S* in series

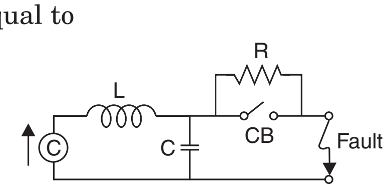
(*c*) *P* and *T* in series (*d*) *P* and *Q* in parallel.

340. One current transformer (CT) is mounted over a 3-phase 3-core cable with its sheath and armour removed from the portion covered by the CT. An ammeter placed in the CT secondary would measure

(*a*) the positive sequence current (*b*) the negative sequence current

(*c*) the zero sequence current (*d*) three times the zero sequence.

341. In connection with the arc extinction in circuit breaker, resistance switching is employed wherein a resistance is placed in parallel with the poles of the circuit breaker as shown in figure. This process introduces damping in the LC circuit. For critical damping, the value of 'R' should be equal to



(*a*) $\sqrt{\dfrac{C}{L}}$ (*b*) $0.5\sqrt{\dfrac{C}{L}}$

(*c*) $0.5\sqrt{\dfrac{L}{C}}$ (*d*) $\dfrac{1}{2\pi}\sqrt{\dfrac{L}{C}}$.

342. Load frequency control is achieved by properly matching the individual machine's

(*a*) reactive powers (*b*) generated voltages

(*c*) turbine inputs (*d*) turbine and generator ratings.

343. If a voltage-controlled bus is treated as load bus, then which one of the following limits would be violated ?

(*a*) Voltage (*b*) Active power

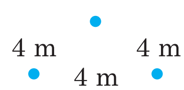
(*c*) Reactive power (*d*) Phase angle.

344. The self-inductance of a long cylindrical conductor due to its internal flux linkages is k H/m. If the diameter of the conductor is doubled, then the self-inductance of the conductor due to its internal flux linkages would be

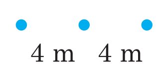
(*a*) 0.5 K H/m (*b*) K H/m

(*c*) 1.414 K H/m (*d*) 4 k H/m.

345. Two arrangements of conductors are proposed for a 3-phase transmission line: one with equilateral spacing of 4 m and the other a flat with 4 m between the conductors as shown in the given figure.



(*i*)



(*ii*)

The conductor diameter in each case is 2 cm. Assuming that the line is transposed in both cases. Which one of the following statements would be true ?

(C_n = capacitance in F/m line to neutral

L = inductance in H/m per phase)

(*a*) $C_{n1} = C_{n2}$ and $L_1 > L_2$ (*b*) $C_{n1} > C_{n2}$ and $L_1 < L_2$

(*c*) $C_{n1} < C_{n2}$ and $L_1 > L_2$ (*d*) $C_{n1} > C_{n2}$ and $L_1 = L_2$.

346. For a given power delivered, if the working voltage of a distributor line is increased to n times, the cross-sectional area A of the distributor line, would be reduced to

(*a*) $\dfrac{1}{n}A$ (*b*) $\dfrac{1}{n^2}A$

(*c*) $\dfrac{1}{2n^2}A$ (*d*) $\dfrac{1}{2n}A$.

347. The following sequence currents were recorded in a power system under a fault condition:

$$I_{\text{positive}} = j\,1.653 \text{ p.u.}$$
$$I_{\text{negative}} = -j\,0.5 \text{ p.u.}$$
$$I_{\text{zero}} = -j\,1.153 \text{ p.u.}$$

The fault is

(*a*) line to ground (*b*) three-phase

(*c*) line to line to ground (*d*) line to line.

348. When bundle conductors are used in place of single conductors, the effective inductance and capacitance will respectively

(*a*) increase and decrease (*b*) decrease and increase

(*c*) decrease and remain unaffected (*d*) remain unaffected and increase.

349. A balanced 3-phase, 3-wire supply feeds balanced star connected resistors. If one of the resistors is disconnected, then the percentage reduction in load will be

(*a*) $33\frac{1}{3}$ (*b*) 50

(*c*) $66\frac{2}{3}$ (*d*) 75.

350. A long overhead transmission line is terminated by its characteristic impedance. Under this operating condition, the ratio of the voltage to the current at different points along the line will

(*a*) progressively increase from the sending-end to the receiving-end

(*b*) progressively increase from the receiving-end to the sending-end

(*c*) remain the same at the two ends, but is higher between the two end being maximum at the centre of the line

(*d*) remain the same at all points.

351. For a transmission line with negligible losses, the lagging reactive power (VAR) delivered at the receiving-end, for a given receiving-end voltage is directly proportional to the

(*a*) square of the line voltage drop (*b*) line voltage drop

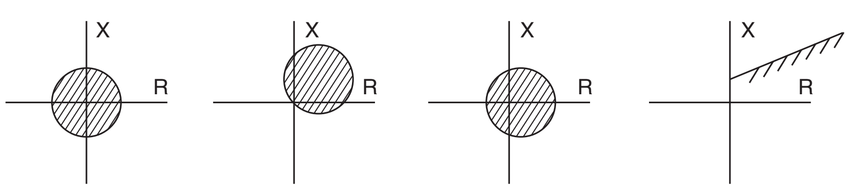
(*c*) line inductive reactance (*d*) line capacitive reactance.

352. Earth wire on EHV overhead transmission line is provided to protect the line against

(*a*) lightning surge (*b*) switching surge

(*c*) excessive fault voltages (*d*) corona effect.

353. Which one of the following is characteristic of an offset MHO relay ?



(a) (b) (c) (d)

354. Consider the following statements about distance relays

1. The effect of an arc may cause the relay to underreach

2. The effect of an intermediate current source may cause relay to overreach

3. The effect of an intermediate load may cause the relay to underreach

4. The presence of dc offset in the fault current may cause the relay to overreach.

Of these statements

(*a*) 2 and 3 are correct (*b*) 1 and 4 are correct

(*c*) 1, 2 and 3 are correct (*d*) 2, 3 and 4 are correct.

355. The inductance of a three-phase transmission line is 1.2 mH/phase/km. If the spacing of conductors and the radius of the conductor are doubled, then the inductance of the line will be

(*a*) 4.8 mH/phase/km (*b*) 1.2 mH/phase/km

(*c*) (1n 2) × 1.2 mH/phase/km (*d*) (1n 4) × 1.2 mH/phase/km.

356. A transmission line has 0.2 pu impedance on a base of 132 kV, 100 MVA. On a base of 220 kV, 50 MVA, it will have a pu impedance of

(*a*) $0.2 \times \frac{50}{100} \times \left(\frac{220}{132}\right)^2$ (*b*) $0.2 \times \frac{100}{50} \times \left(\frac{132}{220}\right)^2$

(*c*) $0.2 \times \frac{50}{100} \times \left(\frac{132}{220}\right)^2$ (*d*) $0.2 \times \frac{100}{50} \times \left(\frac{220}{132}\right)^2$.

357. A three-phase transmission line of negligible resistance and capacitance has an inductive reactance of 100 ohms per phase. When the sending-end and receiving-end voltages are maintained at 110 kV, the maximum power that can be transmitted will be

(*a*) 121 MW (*b*) $121\sqrt{3}$ MW

(*c*) 363 MW (*d*) 363 kW.

358. The per unit "regulation" of a short transmission line having a per-unit resistance voltage drop v_r and a per-unit reactance voltage drop of v_x at rated current and, with power-factor cos θ, is given by

(*a*) $(v_r + v_x) \cos \theta$ (*b*) $(v_r + v_x) \sin \theta$

(*c*) $v_r \cos \theta + v_x \sin \theta$ (*d*) $v_r \sin \theta + v_x \cos \theta$.

359. The surge impedance of a 100 km long underground cable is 50 ohms. The surge impedance of a 40 km long similar cable would be

(*a*) 20 ohms (*b*) 50 ohms

(*c*) 80 ohms (*d*) 125 ohms.

360. A string insulator has 4 units. The voltage across the bottom-most unit is 33.33% of the total voltage. Its string efficiency is

(*a*) 25% (*b*) 33.33%

(*c*) 66.7% (*d*) 75%.

361. The design of insulation for systems above 400 kV, is based upon

(*a*) lightning overvoltage (*b*) switching surges

(*c*) system voltage level (*d*) system load level.

362. Bundled conductors in EHV transmission systems provide

(*a*) increased line reactance (*b*) reduced line capacitance

(*c*) reduced voltage gradient (*d*) increased corona loss.

363. A power system is subjected to a fault which makes the zero sequence component of current equal to zero. The nature of fault is

(*a*) double line to ground fault (*b*) double line fault

(*c*) line of ground fault (*d*) three-phase to ground fault.

364. For which of the following reasons is a differential relay biased to avoid maloperation when used for transformer protection ?

1. Saturation of CTs.
2. Mismatch to CT ratios.
3. Difference in connection of both sides.
4. Current setting multiplier.

Select the correct answer using the codes given below:

(*a*) 1 and 4 (*b*) 1 and 2

(*c*) 2, 3 and 4 (*d*) 1, 2 and 3.

365. Protection scheme used for detection of loss of excitation of a very large generating unit feeding power into a grid employs

(*a*) undervoltage relay (*b*) offset mho relay

(*c*) underfrequency relay (*d*) percentage differential relay.

366. Which one of the following is not true regarding HVDC transmission ?

(*a*) Corona loss is much more in HVDC transmission

(*b*) The power transmission capability of bipolar line is almost the same as that of single-circuit ac line

(*c*) HVDC link can operate between two ac system whose frequencies need not be equal

(*d*) There is no distance limitation for HVDC transmission by underground cable.

367. The incremental cost characteristic of the two units in a plant are given by:

IC_1 = Rs. (0.1 P_1 + 0.8) per MWh

IC_2 = Rs. (0.15 P_2 + 3.0) per MWh

The optimum sharing of load when the total load is 100 MW is

(*a*) P_1 = 60 MW and P_2 = 40 MW (*b*) P_1 = 33.3 MW and P_2 = 66.7 MW

(*c*) P_1 = 40 MW and P_2 = 60 MW (*d*) P_1 = 66.7 MW and P_2 = 33.3 MW.

368. To increase the visual critical voltage of corona for an overhead line, one solid phase-conductor is replaced by a "bundle" of four smaller conductors per phase, having an aggregate cross-sectional area equal to that of the solid conductor. If the radius of the solid conductor is 40 mm, then the radius of each of the bundle conductors would be

(*a*) 10 mm (*b*) 20 mm

(*c*) 28.2 mm (*d*) 30 mm.

369. A single line to ground fault occurs on a three-phase isolated neutral system with a line to neutral voltage of V kV. The potentials on the healthy phases rise to a value equal to

(*a*) $\sqrt{2}$ V kV (*b*) $\sqrt{3}$ V kV

(*c*) 3 V kV (*d*) $\dfrac{1}{\sqrt{3}}$ V kV.

370. In a string of suspension insulators, the voltage distribution across the different units of a string could be made uniform by the use of a grading ring, because it

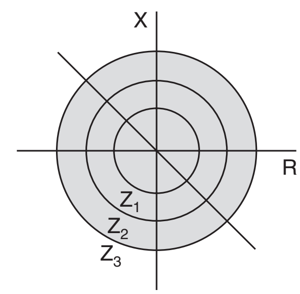
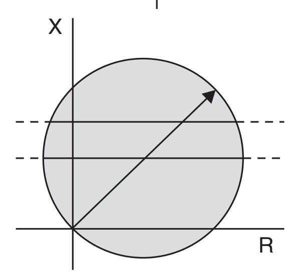
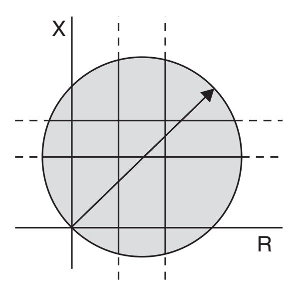
(*a*) forms capacitances with link-pins to cancel the charging current from link-pins

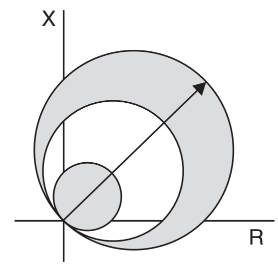
(*b*) forms capacitances which help to cancel the charging current from link-pins

(*c*) increases the capacitances of lower insulator units to cause equal voltage drop

(*d*) decreases the capacitances of upper insulator units to cause equal voltage drop.

371. The following lists relate to the relays and their characteristics for 3-zone protection. Match List I with List II and select the correct answer

List I (Relays)	*List II* (Relay characteristics)
A. Impedance relay	1. 
B. Mho relay	2. 
C. Reactance relay	3. 

4. 

	A	B	C
(a)	1	2	3
(b)	1	4	2
(c)	3	2	1
(d)	4	3	1.

372. A generator is connected to an infinite bus through a double circuit transmission. The fault occuring at the middle of one of the transmission lines is subsequently cleared by opening the circuit breakers at both the ends of the line simultaneously. The transient stability limit of the system is improved by

(*a*) decreasing the excitation of generator

(*b*) decreasing the fault-clearing time

(*c*) increasing the fault-clearing time

(*d*) increasing the transfer reactance between the generator and infinite bus.

373. The velocity of travelling wave through a cable of relative permittivity 9 is

(*a*) 9×10^8 m/s
(*b*) 3×10^8 m/s
(*c*) 10^8 m/s
(*d*) 2×10^8 ms.

374. The bus admittance matrix (Y_{bus}) of a power system is not

(*a*) symmetric
(*b*) a square matrix
(*c*) a full matrix
(*d*) generally having dominant diagonal elements.

375. If the fault current is 3000 A, for an I.D.M.T. relay with a plug setting of 50% and C.T. ratio of 400/5, the plug setting multiplier would be

(*a*) 7.5
(*b*) 15.0
(*c*) 18.75
(*d*) 37.5.

376. A wave-trap is used at the termination of a HVAC overhead line to a station switchyard to

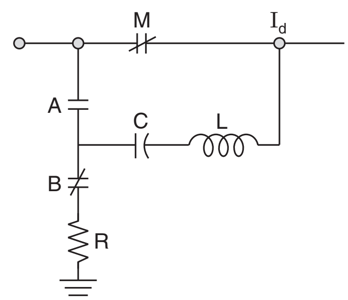
(*a*) prevent the transformer magnetising harmonics from reaching the overhead line

(*b*) damp the incoming surge waves from the overhead lines

(*c*) attenuate sub-harmonic oscillations

(*d*) provide for carrier communication.

377. The given figure shows the schematic diagram of a dc switch (oscillatory discharge type) for interrupting current (I_d) in H.V.D.C. lines, M, A and B are contactors; C, a capacitor; L, an inductor and; R, a high value resistor.



M and B are normally closed contacts, while A is normally open. For interruption of I_d, first B opens and A closes and immediately there-after contact M opens.

The current I_d is interrupted when the resultant current through M

(*a*) is positive-maximum (*b*) is negative-maximum

(*c*) passes through zero (*d*) is without the oscillatory component.

378. A 3-phase overhead transmission line has its conductors horizontally spaced with spacing between adjacent conductors equal to '*d*'. If, now, the conductors of the lines are rearranged to form an equilateral triangle of sides equal to '*d*', then the

(*a*) average capacitance as well as average inductance will increase

(*b*) average capacitance will decrease but the average inductance will increase

(*c*) average capacitance will increase but the average inductance will decrease

(*d*) surge impedance loading of the lines will increase.

379. A 50 Hz 3-phase transmission line of length 100 km has a capacitance of $0.03/\pi$, μ F per km. It is represented as a π model. The shunt admittance at each end of the transmission line will be

(*a*) $150 \times 10^{-6} \angle 90°$ mho (*b*) $100 \times 10^{-6} \angle 90°$ mho

(*c*) $50 \times 10^{-6} \angle 90°$ mho (*d*) $\frac{10^6}{150} \angle 90°$ mho.

380. The insulation resistance of a 20 km long underground cable is 8 mega ohm. Other things being the same, the insulation resistance of a 10 km long cable will be

(*a*) 16 mega ohm (*b*) 32 mega ohm

(*c*) 4 mega ohm (*d*) 2 mega ohm.

381. In an unbalanced 3-phase system, the currents are measured as

$$I_a = \text{zero}, I_b = 6 \angle 60 \text{ and } I_c = 6 \angle -120°$$

The corresponding sequence components will be

	I_{a_0}	I_{a_1}	I_{a_2}
(*a*)	Zero	$3 - j\sqrt{3}$	$-3 + j\sqrt{3}$
(*b*)	Zero	$-3 + j\sqrt{3}$	$3 - j\sqrt{3}$
(*c*)	Zero	$-9 + j3\sqrt{3}$	$9 - j3\sqrt{3}$
(*d*)	Zero	$9 - j3\sqrt{3}$	$-9 + j3\sqrt{3}$.

382. A 3-phase line having negligible resistance and 22 ohm inductive reactance per phase operates with 110 kV sending-end voltage and 100 kV receiving-end voltage. The maximum power that this line can transmit is

(*a*) $500\sqrt{3}$ MW
(*b*) 500 MW
(*c*) 1500 MW
(*d*) $\dfrac{500}{\sqrt{3}}$ MW.

383. A voltage control bus is characterised by the specified

(*a*) real and reactive powers
(*b*) real power and voltage phase angle
(*c*) real power and voltage magnitude
(*d*) reactive power and voltage magnitude.

384. A system is said to be effectively grounded only if R_0/X_1 and X_0/X_1 are respectively (symbols have the usual meanings)

(*a*) ≤ 1 and ≤ 3
(*b*) ≥ 1 and ≥ 3
(*c*) ≤ 2 and ≥ 2
(*d*) ≥ 2 and ≤ 2.

385. Match List I (Coils of a harmonic restrained percentage bias differential relay used for power transformer protection) with List II (Currents flowing through the relay coil) and select the correct answer using the codes given below the lists:

List I	*List II*
A. Operating oil	1. Through current
B. Restraining coil	2. Differential current
C. Bias coil	3. Second harmonic current

Codes:

	A	*B*	*C*
(*a*)	1	3	2
(*b*)	2	3	1
(*c*)	3	2	1
(*d*)	2	1	3

386. A composite transmission and distribution network has lines working at different voltages. Match List I (Voltage level used) with List II (Type of relays used for protection) and select the correct answer using the codes given below the lists:

List I	*List II*
A. 220 kV	1. Directional time lag overcurrent relays
B. 33 kV	2. Carrier current relays
C. 11 kV	3. Distance relays.

Codes:

	A	*B*	*C*
(*a*)	1	3	2
(*b*)	3	2	1
(*c*)	2	1	3
(*d*)	2	3	1

387. Given that

Source voltage $e = E_m \sin(\omega t + \alpha)$, applied at $t = 0$, short-circuit current

$$i = \frac{E_m}{Z} \sin . (\omega t + \alpha - \phi) + K\varepsilon^{-Rt/L}$$

where $\phi = \tan^{-1} \frac{\omega L}{R}$ and K = constant.

A 3-phase short-circuit occurs at the terminals of an alternator having an effective impedance of $Z = \sqrt{R^2 + \omega^2 L^2}$. If the oscillogram of a phase current shows the presence of an exponentially decaying dc component and an ac component decaying to its steady-state value, then the dc component will be zero when

(*a*) $\alpha = \frac{\pi}{2}$

(*b*) $\alpha = \frac{\pi}{2} + \phi$

(*c*) $\alpha = \frac{\pi}{2} - \phi$

(*d*) $\alpha = \phi$.

388. Consider the following statements:

A differential relay is used for a 3-phase transformer protection to avoid maloperation due to

1. saturation of current transformer
2. mismatching of the current ratio for current transformers
3. difference in connections on both sides of power transformer
4. current setting multipliers

Of these statements

(*a*) 1 and 4 are correct (*b*) 1 and 2 are correct

(*c*) 1, 2 and 3 are correct (*d*) 2, 3 and 4 are correct.

389. A dc reactor is connected in series with each pole of a converter station in order to

(*a*) prevent commutation failures in the inverter

(*b*) supply reactive power to the converter

(*c*) improve system stability

(*d*) increase the power transfer capability.

390. In a HVDC transmission, the equivalent circuit representing the operation of a bridge rectifier is as shown in the given figure:

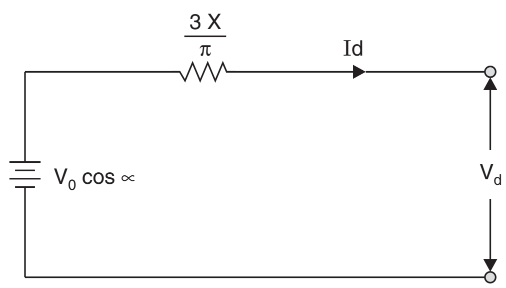
The symbols in the figure have the usual meanings. In this circuit, the voltage drop $\frac{3XI_d}{\pi}$ represents a drop due to

(*a*) ohmic resistance of the thyristors

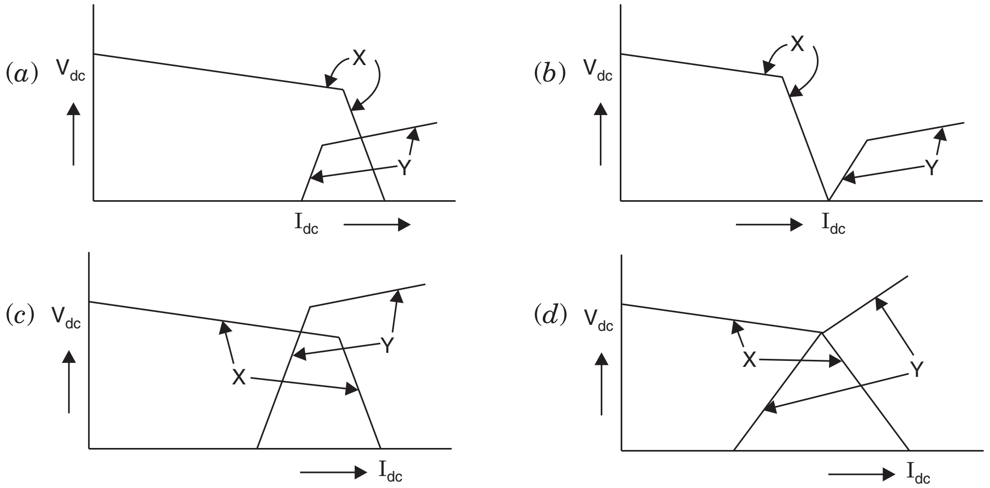
(*b*) ohmic resistance of the rectifier transformer

(*c*) commutation in the rectifier

(*d*) leakage reactance of the transformer.



391. Which one of the following V_{dc} vs I_{dc} characteristics of a HVDC link represents the one that would provide stable steady-state operation ? (in the figures X = rectifier, Y = inverter).



392. The power transmission capability of bipolar lines is approximately
(*a*) half that of 3-phase single circuit line
(*b*) the same as that of 3-phase single circuit line
(*c*) twice that of 3-phase single circuit line
(*d*) thrice that of 3-phase single circuit line.

393. In which one of the following models of transmission lines, is the full charging current assumed to flow over half the length of the line only ?
(*a*) Equivalent-π (*b*) Short line
(*c*) Nominal-π (*d*) Nominal-T.

394. If the receiving-end voltage and current are numerically equal to the corresponding sending-end values, that is, $|V_S| = |V_R|$ and $|I_S| = |I_R|$, then such a line is called
(*a*) an infinite line (*b*) a natural line
(*c*) a tuned line (*d*) a loss-less line.

395. The convergence characteristics of the Netwon-Rophson method for solving a load flow problem is
(*a*) quadratic (*b*) linear
(*c*) geometric (*d*) cubic.

396. If $|V_S| = |V_R| = 66$ KV for three-phase transmission and reactance is 11 ohms/phase, then the maximum power transmission per phase would be
(*a*) 132 MW (*b*) 396 MW
(*c*) 66 MW (*d*) None of the above.

397. For a two-machine system with losses, with the transfer-impedance being resistive, the maximum value of the sending-end power P_{1max} and the maximum receiving-end power P_{2max} will occur at power-angles (δ) in such a manner that
(*a*) both P_{1max} and P_{2max} occur at $\delta < 90°$
(*b*) both P_{1max} and P_{2max} occur at $\delta > 90°$
(*c*) P_{1max} occur at $\delta > 90°$ and P_{2max} at $\delta < 90°$
(*d*) P_{1max} occur at $\delta < 90°$ and P_{2max} at $\delta > 90°$.

398. Consider the following expression:

$$v = f_1 (x - rt) + f_2 (x + rt)$$

where f_1 and f_2 represent two travelling waves on a transmission line. In this case
(*a*) both waves travel in the positive direction of x
(*b*) both waves travel in the negative direction of x
(*c*) wave f_2 travels in the positive direction of x but wave f_1 travels in the negative direction of x
(*d*) wave f_1 travels in the positive direction of x but wave f_2 travels in the negative direction of x.

399. For reducing tower footing resistance, it is better to employ

(*a*) chemical and counterpoise
(*b*) chemical and ground rods
(*c*) ground rods and counterpoise
(*d*) chemical, ground rods and counterpoise.

400. A suspension type insulator has three units with self-capacitance C and ground capacitance of 0.2 C having a string efficiency of

(*a*) 78%
(*b*) 80%
(*c*) 82%
(*d*) 84%.

401. A bundled conductor line compared to a single conductor line (with same conductor cross-sectional area and same mean distance between conductors) has

Self GMD	*Mutual GMD*	*Per phase Inductance*
(*a*) Lower	Nearly same	Higher
(*b*) Higher	Lower	Nearly same
(*c*) Higher	Nearly same	Lower
(*d*) Lower	Higher	Higher.

402. A thyrite lightning arrester has

(*a*) inverse resistance characteristics
(*b*) a gap
(*c*) efficient earthing
(*d*) a combination of inverse resistance characteristics and gap.

403. Differential protection of a generator makes use of the principle that, under normal conditions, the current/currents

(*a*) at the neutral end of a phase winding is zero
(*b*) in each of the phase winding is identical
(*c*) at both ends of the phase winding are equal
(*d*) at the two ends of the phase winding are unequal.

404. The reactance relay is essentially

(*a*) an over-voltage realy with current restraint
(*b*) an over-voltage relay with directional restraint
(*c*) a directional relay with voltage restraint
(*d*) a directional relay with current restraint.

405. Severe over-voltages are produced during arcing faults in a power system with the neutral

(*a*) isolated
(*b*) solidly earthed
(*c*) earthed through a low resistance
(*d*) earthed through an inductive coil.

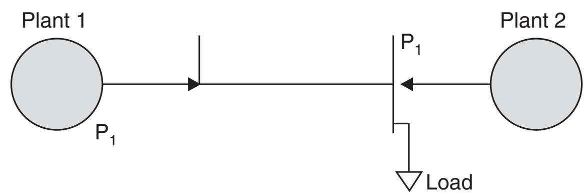
406. In HVDC transmission system, rectifier firing angle alpha is kept near

(*a*) 0°
(*b*) 15°
(*c*) 30°
(*d*) 90°.

407. A two-bus system is shown in the given figure. When 100 MW is transmitted from plant 1 to the load, the transmission loss is 10 MW. The incremental fuel costs of the two plants are

$\frac{dC_1}{dP_1} = 0.02\,P_1 + 16$ and $\frac{dC_2}{dP_2} = 0.04\,P_2 + 20$

The optimum distribution of total load of 260 MW between the two plants when losses are included but not coordinated is



P_1	P_2
(*a*) 300 MW	50 MW
(*b*) 240 MW	20 MW
(*c*) 13 MW	130 MW
(*d*) 220 MW	60 MW.

408. The number of discs in a string of insulators for 400 KV ac overhead line lies in the range of

(*a*) 32 to 33 (*b*) 22 to 23
(*c*) 15 to 16 (*d*) 9 to 10.

409. The inertia constant of a 100 MVA, 50 Hz, 4-pole generator is, 10 MJ/MVA. If the mechanical input to the machine is suddenly raised from 50 MW to 75 MW, the rotor acceleration will be equal to

(*a*) 225 electrical degree/s^2 (*b*) 22.5 electrical degree/s^2
(*c*) 125 electrical degree/s^2 (*d*) 12.5 electrical degree/s^2.

410. Insulation coordination for UHV lines (above 500 kV) is done based on

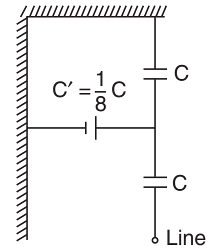
(*a*) lightning surges (*b*) lightning surges and switching surges
(*c*) switching surges (*d*) none of the above.

411. The number of discs in a string of insulators for 220 kV ac overhead transmission line lies in the range of

(*a*) 22 to 25 (*b*) 20 to 21
(*c*) 15 to 16 (*d*) 9 to 10.

412. The insulation resistance of a single-core cable is 200 MΩ/km. The insulation resistance for 5 km length is

(*a*) 40 MΩ (*b*) 1000 MΩ
(*c*) 200 MΩ (*d*) 8 MΩ.



413. The equivalent capacitor arrangement of two string insulator is shown in the above figure. The maximum voltage that each unit can withstand should not exceed 17.5 kV. The line-voltage of the complete string is

(*a*) 17.5 kV (*b*) 33 kV

(*c*) 35 kV (*d*) 37.3 kV.

414. Consider the following statements :

1. By using bundle conductors in an overhead line, the corona loss is reduced.
2. By using bundle conductors, the inductance of transmission line increases and capacitance reduces.
3. Corona loss causes interference in adjoining communication lines.

Which of these statements are correct ?

(*a*) 1 and 2 (*b*) 2 and 3

(*c*) 1 and 3 (*d*) 1, 2 and 3.

415. Which one of the following sequences of operations represents the rated operating duty cycle of a circuit breaker ?

(*O*-open ; *C*-close ; t = 3 sec., T = 3 min)

(*a*) *O–t–CO–T–CO* (*b*) *O–CO–t–CO–T–C*

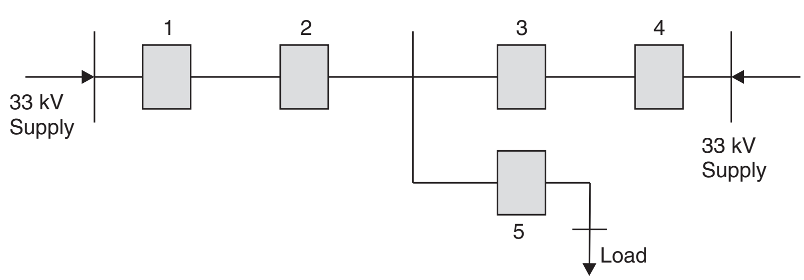
(*c*) *O–C–t–OC–T* (*d*) *O–CO–T–CO–T–C*.

416. Four alternators, each rated at 5 MVA, 11 kV with 20% reactance are working in parallel. The short-circuit level at bus bars is

(*a*) 6.25 MVA (*b*) 20 MVA

(*c*) 25 MVA (*d*) 100 MVA.

417. The distribution system shown in the figure is to be protected by over current system of protection. For proper fault discrimination, directional overcurrent relays will be required at locations



(*a*) 1 and 4 (*b*) 2 and 3

(*c*) 1, 4 and 5 (*d*) 2, 3 and 5.

418. If the fault current is 2 kA, the relay setting is 50% and the CT ratio is 400/5, then plug setting multiplier will be

(*a*) 25 (*b*) 15

(*c*) 50 (*d*) 12.5.

419. A three-phase 11/66 kV, delta/star transformer, protected by Merz-price scheme has CT ratio of 400/5 on L.T side. Ratio of C.T. on H.T. side will be equal to

(*a*) 1 : 23 (*b*) 23 : 1

(*c*) 23 : $\sqrt{3}$ (*d*) $\sqrt{3}$: 23.

420. If, for a given alternator in economic operation mode, the incremental cost is given by (0.012P + 8) Rs./mwh, $\frac{dP_L}{dP}$ = 0.2 and plant λ = 25, then the power generation is

(*a*) 1000 MW (*b*) 1250 MW

(*c*) 750 MW (*d*) 1500 MW.

421. Two generators rated at 200 MW and 400 MW are operating in parallel. Both the governors have a drop of 4%, when the total load is 300 MW. They share the load as (suffix '1' is used for generator 200 MW and suffix '2' is used for generator 400 MW)

(*a*) P_1 = 100 MW and P_2 = 200 MW (*b*) P_1 = 150 MW and P_2 = 150 MW

(*c*) P_1 = 200 MW and P_2 = 100 MW (*d*) P_1 = 200 MW and P_2 = 400 MW.

422. The impedance per phase of 3-phase transmission line on a base of 100 MVA, 100 kV is 2 pu. The value of this impedance on a base of 400 MVA and 400 kV would be

(*a*) 1.5 pu (*b*) 1.0 pu

(*c*) 0.5 pu (*d*) 0.25 pu.

423. The values of A, B, C and D constants for a short transmission line are respectively

(*a*) Z, 0, 1 and 1 (*b*) 0, 1, 1 and 1

(*c*) 1, Z, 0 and 1 (*d*) 1, 1, Z and 0.

424. If a short transmission line is delivering to a lagging *pf* load, the sending-end *pf* would be (notations have their usual meaning)

(*a*) $\frac{V_R \cos\phi + IR \sin\phi}{V_S}$ (*b*) $\frac{V_R \cos\phi + IR}{V_S}$

(*c*) $\frac{V_R \sin\phi + IR}{V_S}$ (*d*) $\frac{V_R \sin\phi + IR \cos\phi}{V_S}$.

425. If the positive, negative and zero sequence reactances of an element of a power system are 0.3, 0.3 and 0.8 respectively, then the element would be a

(*a*) synchronous generator (*b*) synchronous motor

(*c*) static load (*d*) transmission line.

426. An isolated synchronous generator with tansient reactance equal to 0.1 pu on a 100 MVA base is connected to the high voltage bus through a step up transformer of reactance 0.1 pu on 100 MVA base. The fault level at the bus is

(*a*) 1000 MVA (*b*) 500 MVA

(*c*) 50 MVA (*d*) 80 MVA.

427. If a generator of 250 MVA rating has an inertia constant of 6 MJ/MVA, its inertia constant on 100 MVA base is

(*a*) 15 MJ/MVA (*b*) 10.5 MJ/MVA
(*c*) 6 MJ/MVA (*d*) 2.4 MJ/MVA.

428. An overhead line with surge impedance of 400 Ω is terminated through a resistance 'R'. A surge travelling over the line will not suffer any reflection at the junction, if the value of R is
(*a*) 100 Ω (*b*) 200 Ω
(*c*) 400 Ω (*d*) 800 Ω.

429. An overhead line with a surge impedance of 400 Ω is connected to a transformer by a short length of cable of surge impedance 100 Ω. If a rectangular surge wave of 40 kV travels along the line towards the cable, then the voltage of the wave travelling from the junction of the overhead line through the cable towards the transformer would be
(*a*) 16 kV (*b*) 32 kV
(*c*) 30 kV (*d*) 36 kV.

430. Disruptive corona begins in smooth cylindrical conductors in air at NTP if the electric field intensity at the conductor surface goes up to
(*a*) 21.1 kV(rms)/cm (*b*) 21.1 kV(peak)/cm
(*c*) 21.1 kV(average)/cm (*d*) 21.1 kV(rms)/m.

431. In Merz Price percentage differential protection of a $\Delta - Y$ transformer, the *CT* secondaries connection in the primary and secondary windings of the transformer would be in the form of
(*a*) $\Delta - Y$ (*b*) $Y - \Delta$
(*c*) $\Delta - \Delta$ (*d*) $Y - Y$.

432. When there is interference in an overhead communication line running parallel and in close proximity to an overhead power line, the voltage induced in the communication line in the longitudinal and lateral directions by the power line are due to
(*a*) magnetic induction and electric induction respectively
(*b*) electric induction and magnetic induction respectively
(*c*) both magnetic induction and electric induction
(*d*) magnetic induction only.

433. In the case of a HVDC system, there is
(*a*) charging current but no skin effect (*b*) no charging current but skin effect
(*c*) neither charging current nor skin effect
(*d*) both charging current and skin effect.

434. Shunt compensation in an EHV line is used to improve
(*a*) stability and fault level (*b*) fault level and voltage profile
(*c*) voltage profile and stability (*d*) stability, fault level and voltage profile.

435. The cost function of a 50 MW generator is given by (P_i is the generator loading)

$$F(P_i) = 225 + 53\,P_i + 0.02\,P_i^2$$

When 100% loading is applied, the incremental fuel cost (IFC) will be

(*a*) Rs. 55 per MWh (*b*) Rs. 55 per MW

(*c*) Rs. 33 per MWh (*d*) Rs. 33 per MW.

436. In terms of power generation and B_{mn} coefficients, the transmission loss for a two-plant system is (Notations have their usual meaning)

(*a*) $P_1^2B_{11} + 2P_1P_2B_{12} + P_2^2B_{22}$ (*b*) $P_1^2B_{11} - 2P_1P_2B_{12} + P_2^2B_{22}$

(*c*) $P_2^2B_{11} + 2P_1P_2B_{12} + P_1^2B_{22}$ (*d*) $P_1^2B_{11} + P_1P_2B_{12} + P_2^2B_{22}$.

437. The *ABCD* constants of a 3-phase transmission line are

$$A = D = 0.8 \angle 1° \; ; B = 170 \angle 85° \; ; C = 0.002 \angle 90.4° \text{ mho}$$

The sending end voltage is 400 kV. The receiving end voltage under no load condition is

(*a*) 400 kV (*b*) 500 kV

(*c*) 320 kV (*d*) 417 kV.

438. In a short transmission line, voltage regulation is zero when the power factor angle of the load at the receiving end side is equal to

(*a*) $\tan^{-1}\left(\frac{X}{R}\right)$ (*b*) $\tan^{-1}\left(\frac{R}{X}\right)$

(*c*) $\tan^{-1}\left(\frac{X}{Z}\right)$ (*d*) $\tan^{-1}\left(\frac{R}{Z}\right)$.

439. Consider the following statements :

Surge impedance loading of a transmission line can be increased by

1. increasing its voltage level.
2. addition of lumped inductance in parallel.
3. addition of lumped capacitance in series.
4. reducing the length of the line.

Of these statements

(*a*) 1 and 3 are correct (*b*) 1 and 4 are correct

(*c*) 2 and 4 are correct (*d*) 3 and 4 are correct.

440. The load currents in short circuit calculation are neglected because

1. Short circuit currents are much larger than load currents.
2. Short circuit currents are greatly out of phase with load currents.

Which of these statement(s) is/are correct ?

(*a*) Neither 1 nor 2 (*b*) 2 alone

(*c*) 1 alone (*d*) 1 and 2.

441. The surge impedance of a 3-phase, 400 kV transmission line is 400 Ω. The surge impedance loading (SIL) is

(*a*) 400 MW (*b*) 100 MW

(*c*) 1600 MW (*d*) 200 MW.

442. If a travelling-wave travelling along a loss-free overhead line does not result in any reflection after it has reached the far end, then the far end of the line is

(*a*) open circuited (*b*) short circuited

(*c*) terminated into a resistance equal to surge impedance of the line

(*d*) terminated into a capacitor.

443. A travelling wave 400/1/50 means crest value of

(*a*) 400 V with rise time of 1/50 s

(*b*) 400 kV with rise time 1 s and fall time 50 s

(*c*) 400 kV with rise time 1 μs and fall time 50 μs

(*d*) 400 MV with rise time 1 μs and fall time 50 μs.

444. If a 500 MVA, 11 kV three-phase generator at 50 Hz feeds, through a transfer impedance of (0.0 + J 0.605)Ω per phase, an infinite bus also at 11 kV ; then the maximum steady state power transfer on the base of 500 MVA and 11 kV is

(*a*) 1.0 pu (*b*) 0.8 pu

(*c*) 0.5 pu (*d*) 0.4 pu.

445. Installation of capacitors at suitable locations and of optimum size in a distribution system results in

1. improved voltage regulation.
2. reduction in distribution power losses.
3. reduction of kVA rating of distribution transformers.

Select the correct answer from the codes given below :

(*a*) 1 alone (*b*) 1 and 2

(*c*) 1, 2 and 3 (*d*) 3 alone

446. In a three unit insulator string, voltage across the lowest unit is 17.5 kV and string efficiency is 84.28%. The total voltage across the string will be equal to

(*a*) 8.825 kV (*b*) 44.25 kV

(*c*) 88.25 kV (*d*) 442.5 kV.

447. Bundled conductors are used for EHV transmission lines primarily for reducing the

(*a*) corona loss (*b*) surge impedance of the line

(*c*) voltage drop across the line (*d*) I^2R losses.

448. The good effect of corona on overhead lines is to

(*a*) increase the line carrying capacity due to conducting ionised air envelop around the conductor

(*b*) increase the power factor due to corona loss

(*c*) reduce the radio interference from the conductor

(*d*) reduce the steepness of surge fronts.

449. The principal information obtained from load flow studies in a power system are

1. magnitude and phase angle of the voltage at each bus.
2. reactive and real power flows in each of lines.

3. total power loss in the network.

4. transient stability limit of the system.

Select the correct answer from the codes given below :

(*a*) 1 and 2 (*b*) 3 and 4

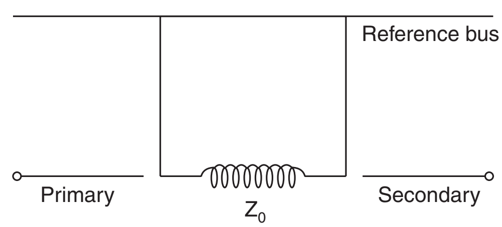
(*c*) 1, 2 and 3 (*d*) 2 and 4.

450. If all the sequence voltages at the fault point in a power system are equal, then the fault is a

(*a*) three-phase fault (*b*) line to ground fault

(*c*) line to line fault (*d*) double line to ground fault.

451.



A three-phase transformer having zero-sequence impedance of Z_0 has the zero-sequence network as shown in the above figure. The connections of its windings are

(*a*) star-star (*b*) delta-delta

(*c*) star-delta (*d*) delta-star with neutral grounded.

452. Which one of the following relays has the capability of anticipating the possible major fault in a transformer ?

(*a*) Overcurrent relay (*b*) Differential relay

(*c*) Buchholz relay (*d*) Overfluxing relay.

453. In a 220 kV system, the inductance and capacitance up to the circuit breaker location are 25 mH and 0.025 μF respectively. The value of resistor required to be connected across the breaker contacts which will give no transient oscillations, is

(*a*) 25 Ω (*b*) 250 Ω

(c) 500 Ω (*d*) 1000 Ω.

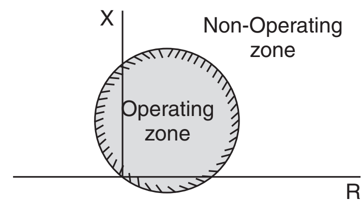
454. Match List I (Equipments) with List II (Applications) and select the correct answer using the codes given below the lists :

List I (*Equipments*)	List II (*Applications*)
A. Metal oxide arrester	1. Protects generator against short circuit faults
B. Isolator	2. Improves transient stability
C. Auto-reclosing *C.B.*	3. Allow *C.B.* maintenance.
D. Differential relay	4. Provides protection against surges

Codes :

(*a*)	*A*	*B*	*C*	*D*	(*b*)	*A*	*B*	*C*	*D*
	4	3	2	1		3	4	1	2
(*c*)	*A*	*B*	*C*	*D*	(*d*)	*A*	*B*	*C*	*D*
	4	3	1	2		3	4	2	1

455. The operating characteristic of a distance relay in the *R*-*X* plane is shown in the given figure. It represents operating characteristic of a



(*a*) reactance relay

(*b*) directional impedance relay

(*c*) impedance relay

(*d*) mho relay.

456. The bus admittance matrix of a power system is given as

$$\begin{array}{c} \\ 1 \\ 2 \\ 3 \end{array}\begin{array}{c} \begin{array}{ccc} \quad 1 & \quad\quad 2 & \quad\quad 3 \end{array} \\ \begin{bmatrix} -j\,50 & +j\,10 & +j\,5 \\ +j\,10 & -j\,30 & +j\,10 \\ +j\,5 & +j\,10 & -j\,25 \end{bmatrix} \end{array}$$

The impedance of line between bus 2 and 3 will be equal to

(*a*) $+j\,0.1$ (*b*) $-j\,0.1$

(*c*) $+j\,0.2$ (*d*) $-j\,0.2$.

457. The component inductance due to the internal flux-linkage of a non-magnetic straight solid circular conductor per meter length, has a constant value, and is independent of the conductor-diameter, because

(*a*) All the internal flux due to a current remains concentrated on the peripheral region of the conductor

(*b*) The internal magnetic flux-density along the radial distance from the centre of the conductor increases proportionately to the current enclosed

(*c*) The entire current is assumed to flow along the conductor axis and the internal flux is distributed uniformly and concentrically

(*d*) The current in the conductor is assumed to be uniformly distributed throughout the conductor cross-section.

458. Match List-I with List-II and select the correct answer using the code given below the Lists :

List-I	*List-II*
A. Graetz Bridge Converter	1. EHV/UHV AC Transmission
B. Series Compensation	2. HVDC Transmission
C. Sag Templates	3. Insulators
D. Grading Ring	4. Tower Location

Codes :

	A	B	C	D		A	B	C	D
(a)	2	1	3	4	(b)	1	2	4	3
(c)	2	1	4	3	(d)	1	2	3	4

459. A rectangular voltage wave is impressed on a loss-free overhead line, with the far end of the line being short-circuited. On reaching the end of this line

(*a*) The current wave is reflected back with positive sign, but the voltage wave with negative sign

(*b*) The current wave is reflected back with negative sign, but the voltage wave with positive sign

(*c*) Both the current and the voltage waves are reflected with positive sign

(*d*) Both the current and the voltage waves are reflected with negative sign.

460. Two insulator discs of identical capacitance value C make up a string for a 22 kV, 50 Hz, single-phase overhead line insulation system. If the pin to earth capacitance is also C, then the string efficiency is

(*a*) 50% (*b*) 75%

(*c*) 90% (*d*) 86%.

461. Which one of the following statements is not correct for the use of bundled conductors in transmission lines ?

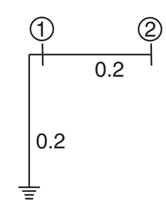
(*a*) Control of voltage gradient

(*b*) Reduction in corona loss

(*c*) Reduction in radio interference

(*d*) Increase in interference with communication lines.

462. In the network as shown here, the marked parameters are p.u. impedances. The bus-admittance matrix of the network is



(*a*) $\begin{bmatrix} 10 & -5 \\ -5 & 5 \end{bmatrix}$ (*b*) $\begin{bmatrix} 5 & -5 \\ -5 & 10 \end{bmatrix}$

(*c*) $\begin{bmatrix} -10 & 5 \\ 5 & -5 \end{bmatrix}$ (*d*) $\begin{bmatrix} -5 & 5 \\ 5 & -10 \end{bmatrix}$.

463. Consider the following statements with reference to protective relays :

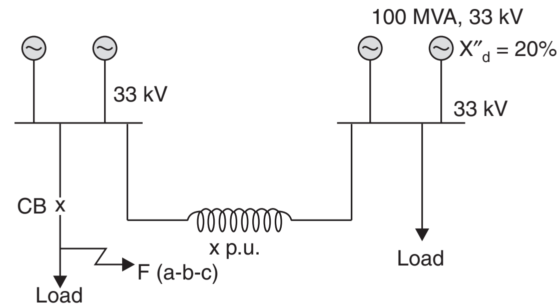
1. The minimum relay coil current at which the relay operates is called pick-up value
2. The pick-up value of a relay is 7.5 *A* and fault current is 30 *A*. Therefore, its plug setting multiplier is 5
3. An earth fault current is generally lesser than the short-circuit current
4. Induction relays are used with both a.c. and d.c. quantities.

Which of these statements are correct ?

(*a*) 1 and 2 (*b*) 2 and 3

(*c*) 1 and 3 (*d*) 1, 2 and 4.

464. Four identical 100 MVA, 33 kV generators are operating in parallel, as shown above, in two bus-bar sections, interconnected through a current limiting reactor of x p.u. reactance on the generator-base. Each generator has a reactance of 0.2 p.u.



The value of the reactor x to limit a symmetrical short-circuit (a-b-c) current through the circuit breaker to 1500 MVA is

(*a*) 0.05 p.u. (*b*) 0.10 p.u.

(*c*) 0.15 p.u. (*d*) 0.20 p.u.

465. The zero sequence current of a generator for line to ground fault is j 2.4 p.u. Then the current through the neutral during the fault is

(*a*) j 2.4 p.u. (*b*) j 0.8 p.u.

(*c*) j 7.2 p.u. (*d*) j 0.24 p.u.

466. For a fixed receiving end and sending end voltages in a transmission system, what is the locus of the constant power ?

(*a*) A straight line (*b*) An ellipse

(*c*) A parabola (*d*) A circle.

467. At slack bus, which one of the following combinations of variables is specified ?

(*a*) $|V|, \delta$ (*b*) P, Q

(*c*) $P, |V|$ (*d*) $Q, |V|$.

(The symbols have their usual meaning)

468. Consider the following quantities :

1. Real power 2. Reactive power
3. Power factor 4. Input current
5. Bus voltage magnitude 6. Bus voltage phase-angle.

For the purpose of the load flow studies of a power system, each bus or node is associated with which one of the combinations of the above four quantities ?

(*a*) 1, 3, 4 and 5 (*b*) 1, 2, 3 and 4

(*c*) 2, 3, 5 and 6 (*d*) 1, 2, 5 and 6.

469. Normally Z_{BUS} matrix is a

(*a*) Null matrix (*b*) Sparse matrix

(*c*) Full matrix (*d*) Unity matrix.

470. If $\alpha = e^{j\frac{2\pi}{3}}$, and $I = AI_S$ where I is equal to phase current vector, and I_S is equal to symmetrical current vector, then which one of the following matrices is the symmetrical components transformation matrix A ?

(*a*) $\begin{bmatrix} 1 & 1 & 1 \\ 1 & \alpha & \alpha^2 \\ 1 & \alpha^2 & \alpha \end{bmatrix}$ (*b*) $\begin{bmatrix} 1 & \alpha & \alpha^2 \\ 1 & 1 & 1 \\ 1 & \alpha^2 & \alpha \end{bmatrix}$

(*c*) $\begin{bmatrix} 1 & 1 & 1 \\ 1 & \alpha^2 & \alpha \\ 1 & \alpha & \alpha^2 \end{bmatrix}$ (*d*) $\begin{bmatrix} 1 & \alpha^2 & \alpha \\ 1 & \alpha & \alpha^2 \\ 1 & 1 & \alpha \end{bmatrix}$.

471. $Z_{pu}{}^{old}$ is the per unit impedance on the power base $S_B{}^{old}$ and voltage base $V_B{}^{old}$. What would be the per unit impedance on the new power base $S_B{}^{\text{new}}$ and voltage base $V_B{}^{new}$?

(*a*) $Z_{pu}{}^{new} = Z_{pu}{}^{old} \dfrac{S_B{}^{old}}{S_B{}^{new}} \left(\dfrac{V_B{}^{new}}{V_B{}^{old}}\right)^2$ (*b*) $Z_{pu}{}^{new} = Z_{pu}{}^{old} \dfrac{S_B{}^{new}}{S_B{}^{old}} \left(\dfrac{V_B{}^{old}}{V_B{}^{new}}\right)^2$

(*c*) $Z_{pu}{}^{new} = Z_{pu}{}^{old} \dfrac{S_B{}^{new}}{S_B{}^{old}} \left(\dfrac{V_B{}^{new}}{V_B{}^{old}}\right)$ (*d*) $Z_{pu}{}^{new} = Z_{pu}{}^{old} \dfrac{S_B{}^{old}}{S_B{}^{new}} \left(\dfrac{V_B{}^{new}}{V_B{}^{old}}\right)$.

472. Consider the following statements regarding the fault analysis :

1. The neutral grounding impedance Z_n appears as $3Z_n$ in zero sequence equivalent circuit.
2. For faults on transmission lines, 3-phase fault is the least severe amongst other faults.
3. The positive and negative sequence networks are not affected by method of neutral grounding.

Which of the statements given above are correct ?

(*a*) 2 and 3 (*b*) 1 and 2

(*c*) 1 and 3 (*d*) 1, 2 and 3.

473. If a sudden short circuit occurs on a power system (considered as *R-L* series circuit), the current wave-form consists of

1. a decaying a.c. current.
2. a decaying d.c. current.

Let the alternator reactance be X and the power system resistance R. Which one of the following is correct ?

(*a*) The decay in (1) is caused by the increase in X but in (2) is caused by R

(*b*) The decay in (1) is caused by R but in (2) is caused by increase in X

(*c*) The decay in both (1) and (2) is caused by R

(*d*) The decay in both (1) and (2) is caused by the increase in X.

474. Match List I (Types of Relays) with List II (Types of Protection) and select the correct answer using the codes given below the lists :

List I (Types of Relyas)	List II (Types of Protection)
A. Directional relay	1. Relay operates for fault within certain distance of its location
B. Impedance relay	2. Relay will trip for fault in one location and block for all other locations

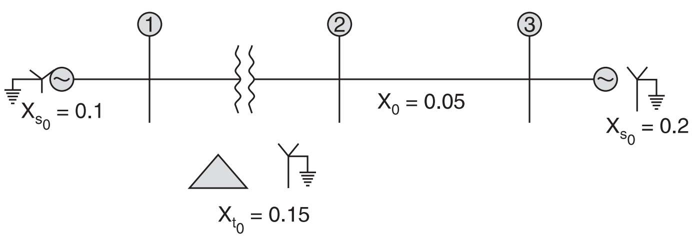
C. Differential relay — 3. High speed protection for entire transmission line

D. Pilot relay — 4. The principle of current continuity is used to devise a simple and effective relaying system over a small physical space.

Codes :

	A	B	C	D		A	B	C	D
(*a*)	1	2	4	3	(*b*)	2	1	3	4
(*c*)	2	1	4	3	(*d*)	1	2	3	4

475. The zero squence reactances (in p.u.) are indicated in the network shown in the figure. The zero sequence driving-point reactance of node 3 will be



(*a*) 0.05 (*b*) 0.1

(*c*) 0.2 (*d*) 0.3.

476. The *A*, *B*, *C* constants of a nominal *T*-circuit are given as

$$A = .95 \angle 0.5° \ ; B = 85 \angle 70° \text{ ohm} \ ; C = 0.0005 \angle 90° \text{ ohm}$$

If the two terminal voltages are held constant at 66 kV, then the steady-state stability limit of the line will be

(*a*) 0.3312 MW (*b*) 3.312 MW

(*c*) 33.12 MW (*d*) 331.2 MW.

477. A feeder with reactance of 0.2 p.u. has a sending-end voltage of 1.2 p.u. If the reactive power supplied at the receiving end of the feeder is 0.3 p.u., the approximate voltage drop in the feeder will be

(*a*) 0.042 p.u. (*b*) 0.052 p.u.

(*c*) 0.060 p.u. (*d*) 0.070 p.u.

478. Which one of the following matrices reveals the topology of the power system network ?

(*a*) Bus incidence matrix (*b*) Primitive impedance matrix

(*c*) Primitive admittance matrix (*d*) Bus impedance matrix.

479. The equivalent Thevenins bus admittance matrix of a two-bus system with identical generators on both buses is $\begin{bmatrix} -j30 & +j10 \\ +j10 & -j30 \end{bmatrix}$

The generator reactance and interconnecting line reactance will be respectively

(*a*) $j0.05$ and $j0.1$ (*b*) $-j0.05$ and $j0.1$

(*c*) $-j0.05$ and $-j0.1$ (*d*) $j0.1$ and $j0.05$.

480. A fault occurring on an end-supplied transmission line is more severe from the point of view of RRRV if it is a

(*a*) long line fault (*b*) short line fault
(*c*) medium line fault (*d*) generator fault.

481. The insulation strength of EHV lines is mainly governed by

(*a*) switching overvoltages (*b*) lightning overvoltages
(*c*) power frequency overvoltages (*d*) dynamic overvoltages.

482. The coefficient of reflection for current of an open-ended line is

(*a*) – 1 (*b*) 1
(*c*) 0.5 (*d*) zero.

483. A relay most likely to operate undesirably on power swings is the

(*a*) over-current relay (*b*) inverse definite minimum time relay
(*c*) mho relay (*d*) reactance relay.

484. Which of the following reasons make HRC fuses preferable in power circuits ?

1. They have inverse time characteristics.
2. They are consistent in performance.
3. They can be selected for proper discrimination.

Select the correct answer by using the codes given below :

Codes :

(*a*) 1 and 2 (*b*) 2 and 3
(*c*) 1 and 3 (*d*) 1, 2 and 3.

485. Consider the following statements about the utility of delta-connected tertiary windings in star-star connected power transformer :

1. It makes supply available for small loads.
2. It provides low reactance paths for zero sequence currents.
3. It is used to suppress harmonic voltages.
4. It is used to allow flow of earth fault current for operation of protective relays.

Which of these statements are correct ?

(*a*) 1 and 2 (*b*) 2, 3 and 4
(*c*) 1, 3 and 4 (*d*) 1, 2, 3 and 4.

486. A 3-phase transmission line conductors were arranged in horizontal spacing, with '*d*' as the distance between adjacent conductors. If these conductors are rearranged to form an equilateral triangle with sides equal '*d*', then the

(*a*) capacitance and the inductance will increase
(*b*) capacitance will increase and the inductance will decrease
(*c*) capacitance and the inductance will remain the same
(*d*) capacitance will decrease and the inductance will increase.

487. The inductance per unit length of an overhead line due to internal flux linkages

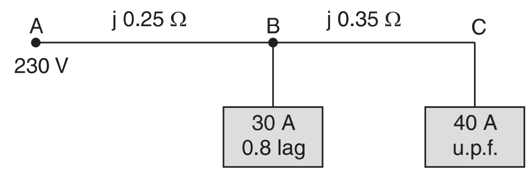
(*a*) depends on the size of the conductor

(*b*) is independent of the size of conductor and constant

(*c*) depends on the current through the conductor

(*d*) depends on distance between conductors.

488. A single-phase ac distributor supplies two single-phase loads as shown in the given figure. The voltage drop from *A* to *C* is



(*a*) $4.5 + j\ 30$ V

(*b*) $30 + j\ 4.5$ V

(*c*) $4.5\ - j\ 30$ V

(*d*) $j\ 30$ V.

489. A synchronous generator with a synchronous reactance of 1.3 pu is connected to an infinite bus whose voltage is 1 pu, through an equivalent reactance of 0.2 pu. For maximum output of 1.2 pu, the alternator emf must be

(*a*) 1.5 pu (*b*) 1.56 pu

(*c*) 1.8 pu (*d*) 2.5 pu.

490. Consider the following statements :

Transient stability of a synchronous generator feeding power to an infinite bus through a transmission line can be increased by

1. increasing the steam supply to the turbine driving the generator during fault clearing.
2. connecting resistors at the generator terminals during fault condition.
3. employing a faster excitation system.
4. quickly throwing off the generator load.

Which of these statements are correct ?

(*a*) 2 and 3 (*b*) 3 and 4

(*c*) 1 and 2 (*d*) 2 and 4.

491. An overhead transmission line having surge impedance 'Z_1' is terminated to an underground cable of surge impedance 'Z_2'. The reflection coefficient for the travelling wave at the junction of the line and cable is

(*a*) $\dfrac{Z_1 + Z_2}{Z_1 - Z_2}$ (*b*) $\dfrac{Z_2}{Z_1 + Z_2}$

(*c*) $\dfrac{Z_2 - Z_1}{Z_1 + Z_2}$ (*d*) $\dfrac{Z_1 - Z_2}{Z_1 + Z_2}$.

492. Ring main distribution is preferred to a radial system because

(*a*) voltage drop in the feeder is less and supply is more reliable

(*b*) voltage drop in the feeder is less and power factor is high

(*c*) power factor is high and supply is more reliable

(*d*) power factor is high and system is less expensive.

493. If the fault current is 3000 A for a relay with a plug setting of 50% and CT ratio of 1000 : 1, the plug setting multiplier would be

(*a*) 1.5 (*b*) 3

(*c*) 4.5 (*d*) 6.

494. Match List I (Protective schemes) with List II (Equipment) and select the correct answer using the codes given the Lists :

List I	*List II*
A. Mho relays	1. Generators
B. Inverse time overcurrent relays	2. Transmission lines
C. Differential relays	3. Motors

Code :

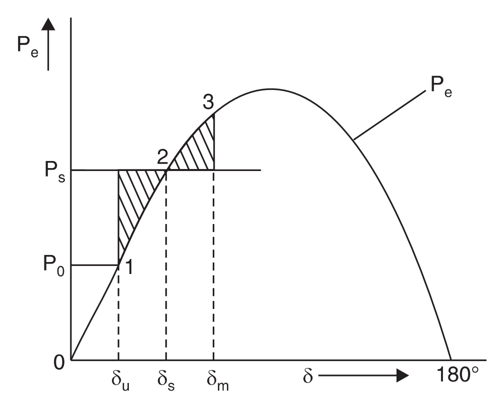
	A	*B*	*C*		*A*	*B*	*C*
(*a*)	2	1	3	(*b*)	2	3	1
(*c*)	3	2	1	(*d*)	1	3	2

495. In terms of plant powers P_n and P_m and loss coefficients B_{mn} the total transmission loss 'P_L' is

(*a*) $\sum_{m=1}^{N} \sum_{n=1}^{N} B_{mn} P_n$ (*b*) $\sum_{m=1}^{N} \sum_{n=1}^{N} P_m B_{mn}$

(*c*) $\sum_{m=1}^{N} \sum_{n=1}^{N} P_m B_{mn} P_n$ (*d*) $\sum_{m=1}^{N} \sum_{n=1}^{N} 2P_m B_{mn}$.

496. The given figure shows electric power input P_e to a lossless synchronous motor as a function of torque angle δ. The load is suddenly increased from P_0 to P_s and the motor oscillates around δ_s between δ_0 and δ_m.



Consider the following statements derived from the figure regarding the relationship between the motor speed ω and its synchronous speed ω_s at different points in the oscillating cycle :

I. At point 1, $\omega = \omega_s$.

II. At point 2, while oscillating from 1 towards 3, $\omega < \omega_s$.

III. At point 3, $\omega < \omega_s$.

IV. At point 2, while oscillating from 3 towards 1, $\omega > \omega_s$.

Which of these statement (s) is/are correct ?

(*a*) II and IV (*b*) I, II and IV

(*c*) I and III (*d*) III alone.

497. The reactance of a generator designated X″ is given as 0.25 pu based on the generator's name plate rating of 18 kV, 500 MVA. If the base for calculations is changed to 20 kV, 100 MVA, the generator reactance X'' on new base will be

(*a*) 1.025 pu (*b*) 0.05 pu

(*c*) 0.0405 pu (*d*) 0.25 pu.

498. Consider the following statements:

The calculations performed using short-line approximate model instead of nomial-π model for a medium length transmission line delivering lagging load at a given receiving-end voltage always result in higher.

1. sending-end current. 2. sending-end power.

3. regulation. 4. Efficiency.

Which of these statements are correct ?

(*a*) 1 and 2 (*b*) 3 and 4

(*c*) 1, 2 and 3 (*d*) 1, 2 and 4.

499. The severity of line to ground and three-phase faults at the terminals of an unloaded synchronous generator is to be same. If the terminal voltage is 1.0 pu, $Z_1 = Z_2 = j0.1$ pu and $Z_0 = j0.05$ pu for the alternator, then the required inductive reactance for neutral grounding is

(*a*) 0.0166 pu (*b*) 0.05 pu

(*c*) 0.1 pu (*d*) 0.15 pu.

500. A short length of cable is connected between dead-end tower and sub-station at the end of a transmission line. Which of the following will decrease, when voltage wave is entering from overhead line to cable ?

1. Velocity of propagation of voltage wave.

2. Steepness of voltage wave.

3. Magnitude of voltage wave.

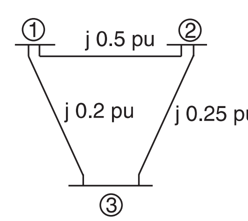
Select the correct answer using the codes given below:

Codes :

(*a*) 1 and 2 (*b*) 2 and 3

(*c*) 3 and 1 (*d*) 1, 2 and 3.

501. A single line diagram of a power system is shown in the given figure. A diagonal elements of the Y_{BUS} matrix is



(*a*) j 0.7 pu, j 0.75 pu, j 0.45 pu

(*b*) $-j$ 0.7 pu., – 0.75 pu., $-j$ 0.45 pu

(*c*) $-j$ 7.0 pu, $-j$ 6.0 pu, $-j$ 9.0 pu

(*d*) j 7.0 pu, $-j$ 6.0 pu, $-j$ 9.0 pu.

502. In load-flow analysis, the load at a bus is represented as

(*a*) a constant current drawn from the bus

(*b*) a constant impedance connected at the bus

(*c*) constant real and reactive powers drawn from the bus

(*d*) a voltage-dependent impedance at the bus.

503. Match List-I (Relay) with List-II (Equipment) and select the correct answer using the codes given below the Lists:

List-I	*List-II*
A. Mho Relay	1. Bus bar
B. Negative Sequence Relay	2. Motor
C. Thermal Relay	3. Generator
	4. Transmission line

Codes :

	A	*B*	*C*		*A*	*B*	*C*
(*a*)	1	3	2	(*b*)	2	4	1
(*c*)	4	2	3	(*d*)	4	3	2

504. Shunt resistor is connected across the contacts of a circuit-breaker in order to

(*a*) damp out the restriking transient (*b*) bypass the arc current

(*c*) limit the short-circuit current

(*d*) reduce the damage to the contacts due to arcing.

505. Maloperation of differential protection of transformers due to magnetizing inrush current is prevented by

(*a*) setting the current of the relay higher than the maximum value of inrush current

(*b*) keeping the time-setting long enough for the inrush current to fall to a value below the primary operating current of the relay

(*c*) bypassing the inrush current from the operating coil of the relay

(*d*) filtering the second harmonic content of the inrush current flowing through the operating coil and passing through the restraining coil.

506. Incremental fuel costs in Rs/MWh for a plant consisting of two units are given by

$$\frac{dF_1}{dP_1} = 0.4P_1 + 400 \text{ and } \frac{dF_2}{dP_2} = 0.48P_2 + 320.$$

The allocation of loads P_1 and P_2 between the units 1 and 2, respectively, for minimum cost of generation for a total load of 900 MW is

(*a*) 200 MW and 700 MW (*b*) 300 MW and 600 MW

(*c*) 400 MW and 500 MW (*d*) 500 MW and 400 MW.

507. For a given base voltage and base volt amperes, the per unit impedance value of an element is x. The per unit impedance value of this element when the voltage and volt amperes bases are both doubled will be

(*a*) $0.5x$ (*b*) x

(*c*) $2x$ (*d*) $4x$.

508. Consider the following statements with respect to an interconnected power system :

1. Frequency will be same at all buses in the system.
2. Voltages can be different at different buses.
3. Both frequency and voltage can be different at different buses.

Which of these statements are correct ?

(*a*) 1, 2 and 3 (*b*) 1 and 3

(*c*) 1 and 2 (*d*) 2 and 3.

509. Consider the following statements :

Overhead transmission lines are provided with earth wires

1. to protect the transmission line from direct lightning strike.
2. to protect the transmission line insulation from the indirect lightning strike.
3. to balance the line currents.
4. to provide path for neutral current.

Which of these statements is/are correct ?

(*a*) 1 and 3 (*b*) 1 and 4

(*c*) 1 only (*d*) 2 only.

510. In Gauss-Seidel method of power flow problem, the number of iterations may be reduced if the correction in voltage at each bus is multiplied by

(*a*) Gauss constant (*b*) Acceleration constant

(*c*) Deceleration constant (*d*) Blocking factor.

511. Consider the following statements :

1. It is easier to construct the Y_{BUS} matrix as compared to Z_{BUS}.
2. Z_{BUS} is a full matrix while Y_{BUS} is sparse.
3. Y_{BUS} can be easily modified whenever the network changes as compared to the Z_{BUS}.

Which of these statements are correct ?

(*a*) 1 and 2 (*b*) 2 and 3

(*c*) 1 and 3 (*d*) 1, 2 and 3.

512. Match List-I (Power system components) with List-II (Relaying schemes) and select the correct answer using the codes given below the Lists :

List-I (*Power system components*)	*List-II* (*Relaying schemes*)
A. Power Transformer	1. Differential relaying
B. Transmission Lines	2. Distance relaying
C. Alternator	

Codes :

	A	*B*	*C*		*A*	*B*	*C*
(*a*)	1	1	2	(*b*)	2	1	1
(*c*)	1	2	1	(*d*)	2	1	2

513. Steady state operating condition of a power system indicates

(*a*) a situation when the connected load is absolutely constant

(*b*) a situation when the generated power is absolutely constant

(c) a situation when both connected load and generated power are equal to each other and remain constant

(d) an equilibrium state around which small fluctuations in power, both in generation and load, occur all the time.

514. Match List-I (*Insulation Type*) with List-II (*Purpose or Configuration*) and select the correct answer using the codes given below the lists :

List-I (*Insulation Type*)	*List-II* (*Purpose or Configuration*)
A. Pin type	1. Low voltage distribution lines
B. Suspension type	2. String of insulators in horizontal position
C. Strain type	3. String of insulators in vertical position
D. Shackle type	4. For voltage upto 33 kV

Codes :

	A	*B*	*C*	*D*		*A*	*B*	*C*	*D*
(*a*)	4	2	3	1	(*b*)	1	3	2	4
(*c*)	4	3	2	1	(*d*)	1	2	3	4

515. Match List-I (*Bus Types*) with List-II (*Pairs of variables*) and select the correct answer using the codes given below the lists :

List-I (*Bus Types*)	*List-II* (*Pairs of variables*)
A. Load bus	1. P and V
B. Generator bus	2. P and Q
C. Slack bus	3. V and δ

Codes :

	A	*B*	*C*		*A*	*B*	*C*
(*a*)	1	2	3	(*b*)	2	3	1
(*c*)	1	3	2	(*d*)	2	1	3

516. Match List-I (*Relays*) with List-II (*Protection*) and select the correct answer using the codes given below the lists :

List-I (*Relays*)	*List-II* (*Protection*)
A. Buchholz relay	1. Feeder
B. Translay relay	2. Transformer
C. Carrier current phase comparison relay	3. Ring main distributor
D. Directional over current relay	4. Long overhead transmission line.

Codes :

	A	*B*	*C*	*D*		*A*	*B*	*C*	*D*
(*a*)	2	3	4	1	(*b*)	4	1	2	3
(*c*)	2	1	4	3	(*d*)	4	3	2	1

517. A three-phase 50 Hz transmission line has a capacitance of line to neutral $C_n = 0.01\ \mu F/km$. The voltage of the line is 100 kV. The charging current per kilometre of line is

(*a*) $\dfrac{314}{\sqrt{3}}$ A/km (*b*) $\dfrac{314}{\sqrt{3} \times 10^3}$ A/km

(*c*) $\dfrac{\sqrt{3} \times 10^3}{314}$ A/km (*d*) $\sqrt{3} \times 10^3$ A/km.

518. A string insulator has 5 units. The voltage across the bottom-most unit is 25% of the total voltage. The string efficiency is

(*a*) 25% (*b*) 50%

(*c*) 80% (*d*) 75%.

519. In load flow studies of a power system, a voltage control bus is specified by

(*a*) Real power and reactive power (*b*) Real power and voltage magnitude

(*c*) Voltage and voltage phase angle (*d*) Reactive power and voltage magnitude.

520. The power angle characteristic of machine-infinite bus system is

$P_c = 2 \sin \delta$ pu

It is operating at $\delta = 30°$. Which one of the following is the synchronising power coefficient at the operating point ?

(*a*) 1.0 (*b*) $\sqrt{3}$

(*c*) 2.0 (*d*) $\dfrac{1}{\sqrt{3}}$.

521. Which methods can be used for the measurement of three phase power for an unbalanced load ?

1. Three voltmeters 2. Two voltmeters and one ammeter
3. Two wattmeters 4. One wattmeter

Select the correct answer using the codes given below :

Codes :

(*a*) 1 only (*b*) 3 only

(*c*) 1 and 3 (*d*) Any one of 1, 2, 3 or 4.

522. Possible faults that may occur on a transmission line are

1. 3-phase fault 2. *L-L-G* fault
3. *L-L* fault 4. *L-G* fault.

The decreasing order of severity of the fault from the stability point of view is :

(*a*) 1-2-3-4 (*b*) 1-4-3-2

(*c*) 1-3-2-4 (*d*) 1-3-4-2.

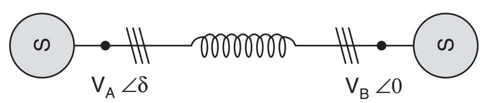
523. Match List I (*Protection Scheme*) with List II (*Type of Fault*) and select the correct answer using the codes given below the lists :

List I (*Protection Scheme*)	*List II* (*Type of Fault*)
A. Differential protection	1. Phase to phase, and phase to ground
B. Buchholz protection	2. Phase to ground
C. Earth fault protection	3. Short circuit
D. Thermal protection	4. Inter-turn
	5. Over loading

Codes :

	A	*B*	*C*	*D*		*A*	*B*	*C*	*D*
(*a*)	1	4	2	5	(*b*)	5	2	3	1
(*c*)	1	3	2	5	(*d*)	5	4	2	1

524. Two identical synchronous machines A and B, running at the same speed, are linked through inductors as shown in the given figure :



Machine A will supply active and reactive power to machine B when δ is

(*a*) positive and V_A is less than V_B

(*b*) positive and V_A is greater than V_B

(*c*) negative and V_A is less than V_B

(*d*) negative and V_A is greater than V_B.

Hint for Problem 287

The general entry in the bus impedance matrix is given by the expression

$$Z_{ij} = \left.\frac{V_i}{I_j}\right|_{I_1 = I_2 = I_3 \ldots = I_n = 0,\, I_j = 1\text{p.u.}}$$

We connect a source of unit current between each node and the reference node. To find out the entries in the row of the corresponding node where the source is connected we obtain value of voltage of various nodes. Since the current is 1 amp. The voltages give the impedance entries at various nodes in the row. For example if we connect a current source between node 1 and the reference. The voltage of node is 1 volt and those of nodes 3 and 2 are w.r.t. reference node, zero volts. Hence entries in the first row are 1 0 0.

Now connect the source between node 2 and reference node. The voltage of node 1 w.r.t. reference is zero that of node 2 is 5 volts and 3 node it is 2 volt, therefore the entries in the second row will be 0 5 2

Similarly it can be seen that when we connect a unity current source between node 3 and reference the entires will be 0 2 2

Therefore option (*c*) is correct.

ANSWERS TO OBJECTIVE QUESTIONS

Question	*Answer*	*Question*	*Answer*	*Question*	*Answer*	*Question*	*Answer*
1.	(*b*)	40.	(*c*)	79.	(*d*)	118.	(*a*)
2.	(*b*)	41.	(*d*)	80.	(*c*)	119.	(*b*)
3.	(*c*)	42.	(*b*)	81.	(*d*)	120.	(*a*)
4.	(*a*)	43.	(*c*)	82.	(*b*)	121.	(*b*)
5.	(*a*)	44.	(*a*)	83.	(*c*)	122.	(*a*)
6.	(*a*)	45.	(*b*)	84.	(*b*)	123.	(*a*)
7.	(*d*)	46.	(*a*)	85.	(*c*)	124.	(*d*)
8.	(*a*)	47.	(*c*)	86.	(*b*)	125.	(*b*)
9.	(*a*)	48.	(*a*)	87.	(*b*)	126.	(*a*)
10.	(*a*)	49.	(*c*)	88.	(*c*)	127.	(*d*)
11.	(*b*)	50.	(*a*)	89.	(*c*)	128.	(*b*)
12.	(*b*)	51.	(*b*)	90.	(*a*)	129.	(*c*)
13.	(*d*)	52.	(*c*)	91.	(*a*)	130.	(*c*)
14.	(*b*)	53.	(*d*)	92.	(*c*)	131.	(*a*)
15.	(*a*)	54.	(*d*)	93.	(*c*)	132.	(*d*)
16.	(*a*)	55.	(*a*)	94.	(*a*)	133.	(*d*)
17.	(*d*)	56.	(*d*)	95.	(*c*)	134.	(*a*)
18.	(*b*)	57.	(*a*)	96.	(*d*)	135.	(*d*)
19.	(*d*)	58.	(*a*)	97.	(*b*)	136.	(*c*)
20.	(*a*)	59.	(*c*)	98.	(*d*)	137.	(*b*)
21.	(*b*)	60.	(*c*)	99.	(*a*)	138.	(*c*)
22.	(*c*)	61.	(*a*)	100.	(*b*)	139.	(*c*)
23.	(*a*)	62.	(*a*)	101.	(*c*)	140.	(*b*)
24.	(*a*)	63.	(*a*)	102.	(*d*)	141.	(*a*)
25.	(*d*)	64.	(*a*)	103.	(*c*)	142.	(*a*)
26.	(*a*)	65.	(*d*)	104.	(*d*)	143.	(*a*)
27.	(*c*)	66.	(*c*)	105.	(*c*)	144.	(*b*)
28.	(*c*)	67.	(*a*)	106.	(*a*)	145.	(*d*)
29.	(*c*)	68.	(*a*)	107.	(*b*)	146.	(*b*)
30.	(*c*)	69.	(*c*)	108.	(*c*)	147.	(*a*)
31.	(*d*)	70.	(*b*)	109.	(*b*)	148.	(*b*)
32.	(*a*)	71.	(*a*)	110.	(*a*)	149.	(*b*)
33.	(*c*)	72.	(*d*)	111.	(*b*)	150.	(*d*)
34.	(*a*)	73.	(*a*)	112.	(*a*)	151.	(*c*)
35.	(*d*)	74.	(*b*)	113.	(*c*)	152.	(*c*)
36.	(*d*)	75.	(*c*)	114.	(*d*)	153.	(*d*)
37.	(*b*)	76.	(*a*)	115.	(*c*)	154.	(*c*)
38.	(*b*)	77.	(*c*)	116.	(*a*)	155.	(*c*)
39.	(*b*)	78.	(*c*)	117.	(*a*)	156.	(*c*)

Question	*Answer*	*Question*	*Answer*	*Question*	*Answer*	*Question*	*Answer*
157.	(*c*)	197.	(*a*)	237.	(*a*)	277.	(*d*)
158.	(*a*)	198.	(*b*)	238.	(*d*)	278.	(*a*)
159.	(*c*)	199.	(*a*)	239.	(*c*)	279.	(*a*)
160.	(*a*)	200.	(*d*)	240.	(*c*)	280.	(*b*)
161.	(*a*)	201.	(*d*)	241.	(*c*)	281.	(*a*)
162.	(*a*)	202.	(*b*)	242.	(*a*)	282.	(*c*)
163.	(*c*)	203.	(*d*)	243.	(*b*)	283.	(*d*)
164.	(*d*)	204.	(*d*)	244.	(*c*)	284.	(*a*)
165.	(*a*)	205.	(*b*)	245.	(*a*)	285.	(*b*)
166.	(*c*)	206.	(*c*)	246.	(c)	286.	(*a*)
167.	(*a*)	207.	(*d*)	247.	(*d*)	287.	(*c*)
168.	(*b*)	208.	(*b*)	248.	(*b*)	288.	(*d*)
169.	(*c*)	209.	(*c*)	249.	(*b*)	289.	(*a*)
170.	(*a*)	210.	(*d*)	250.	(*b*)	290.	(*a*)
171.	(*d*)	211.	(*b*)	251.	(*c*)	291.	(*c*)
172.	(*c*)	212.	(*c*)	252.	(*a*)	292.	(*b*)
173.	(*b*)	213.	(*d*)	253.	(*b*)	293.	(*b*)
174.	(*c*)	214.	(*c*)	254.	(*b*)	294.	(*d*)
175.	(*c*)	215.	(*d*)	255.	(*a*)	295.	(*d*)
176.	(*b*)	216.	(*d*)	256.	(*a*)	296.	(*d*)
177.	(*c*)	217.	(*d*)	257.	(*d*)	297.	(*c*)
178.	(*d*)	218.	(*a*)	258.	(*a*)	298.	(*b*)
179.	(*c*)	219.	(*d*)	259.	(*c*)	299.	(*a*)
180.	(*a*)	220.	(*c*)	260.	(*d*)	300.	(*d*)
181.	(*b*)	221.	(*a*)	261.	(*c*)	301.	(*b*)
182.	(*c*)	222.	(*c*)	262.	(*c*)	302.	(*d*)
183.	(*c*)	223.	(*c*)	263.	(*d*)	303.	(*c*)
184.	(*c*)	224.	(*c*)	264.	(*b*)	304.	(*a*)
185.	(*c*)	225.	(*d*)	265.	(*d*)	305.	(*d*)
186.	(*b*)	226.	(*c*)	266.	(*c*)	306.	(*a*)
187.	(*d*)	227.	(*b*)	267.	(*b*)	307.	(*c*)
188.	(*b*)	228.	(*a*)	268.	(*d*)	308.	(*a*)
189.	(*b*)	229.	(*b*)	269.	(*a*)	309.	(*b*)
190.	(*b*)	230.	(*c*)	270.	(*a*)	310.	(*c*)
191.	(*c*)	231.	(*d*)	271.	(*c*)	311.	(*d*)
192.	(*b*)	232.	(*d*)	272.	(*c*)	312.	(*b*)
193.	(*c*)	233.	(*b*)	273.	(*d*)	313.	(*a*)
194.	(*a*)	234.	(*b*)	274.	(*b*)	314.	(*b*)
195.	(*b*)	235.	(*c*)	275.	(*a*)	315.	(*a*)
196.	(*d*)	236.	(*b*)	276.	(*b*)	316.	(*c*)

Question	*Answer*	*Question*	*Answer*	*Question*	*Answer*	*Question*	*Answer*
317.	(*d*)	357.	(*a*)	397.	(*a*)	437.	(*b*)
318.	(*d*)	358.	(*c*)	398.	(*d*)	438.	(*b*)
319.	(*c*)	359.	(*b*)	399.	(*c*)	439.	(*a*)
320.	(*a*)	360.	(*d*)	400.	(*a*)	440.	(*c*)
321.	(*c*)	361.	(*b*)	401.	(*c*)	441.	(*a*)
322.	(*a*)	362.	(*c*)	402.	(*d*)	442.	(*c*)
323.	(*d*)	363.	(*b*)	403.	(*c*)	443.	(*c*)
324.	(*d*)	364.	(*b*)	404.	(*b*)	444.	(*d*)
325.	(*d*)	365.	(*b*)	405.	(*a*)	445.	(*c*)
326.	(*a*)	366.	(*a*)	406.	(*b*)	446.	(*b*)
327.	(*b*)	367.	(*c*)	407.	(*a*)	447.	(*b*)
328.	(*b*)	368.	(*b*)	408.	(*b*)	448.	(*d*)
329.	(*c*)	369.	(*b*)	409.	(*a*)	449.	(*c*)
330.	(*d*)	370.	(*b*)	410.	(*c*)	450.	(*d*)
331.	(*c*)	371.	(*b*)	411.	(*c*)	451.	(*b*)
332.	(*c*)	372.	(*b*)	412.	(*a*)	452.	(*c*)
333.	(*c*)	373.	(*c*)	413.	(*b*)	453.	(*c*)
334.	(*c*)	374.	(*c*)	414.	(*c*)	454.	(*a*)
335.	(*a*)	375.	(*b*)	415.	(*a*)	455.	(*d*)
336.	(*b*)	376.	(*d*)	416.	(*d*)	456.	(*a*)
337.	(*b*)	377.	(*c*)	417.	(*b*)	457.	(*d*)
338.	(*a*)	378.	(*c*)	418.	(*d*)	458.	(*c*)
339.	(*a*)	379.	(*a*)	419.	(*b*)	459.	(*a*)
340.	(*d*)	380.	(*a*)	420.	(*a*)	460.	(*b*)
341.	(*c*)	381.	(*c*)	421.	(*a*)	461.	(*d*)
342.	(*c*)	382.	(*b*)	422.	(*c*)	462.	(*a*)
343.	(*c*)	383.	(*c*)	423.	(*c*)	463.	(*c*)
344.	(*b*)	384.	(*a*)	424.	(*b*)	464.	(*b*)
345.	(*b*)	385.	(*d*)	425.	(*d*)	465.	(*c*)
346.	(*b*)	386.	(*d*)	426.	(*b*)	466.	(*a*)
347.	(*c*)	387.	(*d*)	427.	(*a*)	467.	(*a*)
348.	(*b*)	388.	(*b*)	428.	(*c*)	468.	(*d*)
349.	(*b*)	389.	(*a*)	429.	(*a*)	469.	(*c*)
350.	(*d*)	390.	(*c*)	430.	(*a*)	470.	(*c*)
351.	(*d*)	391.	(*a*)	431.	(*b*)	471.	(*b*)
352.	(*a*)	392.	(*b*)	432.	(*a*)	472.	(*c*)
353.	(*c*)	393.	(*d*)	433.	(*c*)	473.	(*a*)
354.	(*b*)	394.	(*a*)	434.	(*b*)	474.	(*c*)
355.	(*b*)	395.	(*b*)	435.	(*a*)	475.	(*b*)
356.	(*c*)	396.	(*a*)	436.	(*a*)	476.	(*c*)

Question	*Answer*	*Question*	*Answer*	*Question*	*Answer*	*Question*	*Answer*
477.	(*b*)	489.	(*c*)	501.	(*c*)	513.	(*d*)
478.	(*a*)	490.	(*a*)	502.	(*c*)	514.	(*c*)
479.	(*a*)	491.	(*c*)	503.	(*d*)	515.	(*d*)
480.	(*b*)	492.	(*a*)	504.	(*a*)	516.	(*c*)
481.	(*a*)	493.	(*a*)	505.	(*d*)	517.	(*b*)
482.	(*a*)	494.	(*b*)	506.	(*c*)	518.	(*c*)
483.	(*d*)	495.	(*c*)	507.	(*a*)	519.	(*b*)
484.	(*d*)	496.	(*b*)	508.	(*c*)	520.	(*b*)
485.	(*d*)	497.	(*c*)	509.	(*c*)	521.	(*b*)
486.	(*d*)	498.	(*c*)	510.	(*b*)	522.	(*a*)
487.	(*b*)	499.	(*a*)	511.	(*d*)	523.	(*c*)
488.	(*a*)	500.	(*d*)	512.	(*c*)	524.	(*b*)